Name:____________

GCSE Physics Unit 2 Foundation Tier

Once you have completed a paper and have marked it, fill out the table below, using the grade boundaries provided to add your grade. Then list which questions you did well and which questions you struggled with. For the questions your struggled with, review why and at the reasons. The most common reasons are

- Not understanding basic concepts (e.g. simple definitions)
- Calculation errors (forgetting to square numbers, etc.)
- Not reading questions carefully
- Not answering the full question (describe vs. explain)
- Laboratory/practical setup mistakes

For these questions it is also useful add notes to the question to help you next time and take a photo of these questions. Use these then to guide further revision.

Year	Score /80	Grade	Good Questions	Bad Questions	Reasons
2018					
2019					
2022					
2023					

Grade boundaries -Raw Score

Year	G	F	E	D	C	Max UMS
2018	16	20	24	29	34	39
2019	12	16	20	24	29	34
2022	6	12	18	24	30	36
2023	5	12	19	26	33	40

Surname	Centre Number	Candidate Number
Other Names		0

GCSE – NEW

3420U20-1

S18-3420U20-1

PHYSICS – Unit 2:
Forces, Space and Radioactivity

FOUNDATION TIER

WEDNESDAY, 23 MAY 2018 – AFTERNOON

1 hour 45 minutes

For Examiner's use only		
Question	**Maximum Mark**	**Mark Awarded**
1.	**4**	
2.	**9**	
3.	**7**	
4.	**9**	
5.	**8**	
6.	**7**	
7.	**16**	
8.	**7**	
9.	**13**	
Total	**80**	

3420U201
01

ADDITIONAL MATERIALS

In addition to this paper you will require a calculator and a ruler.

INSTRUCTIONS TO CANDIDATES

Use black ink or black ball-point pen. Do not use gel pen. Do not use correction fluid.

Write your name, centre number and candidate number in the spaces at the top of this page.

Answer **all** questions.

Write your answers in the spaces provided in this booklet. If you run out of space use the additional page at the back of the booklet.

INFORMATION FOR CANDIDATES

The number of marks is given in brackets at the end of each question or part-question.

The assessment of the quality of extended response (QER) will take place in question **7(a)**.

MAY183420U20101

Equations

speed = $\dfrac{\text{distance}}{\text{time}}$	
acceleration [or deceleration] = $\dfrac{\text{change in velocity}}{\text{time}}$	$a = \dfrac{\Delta v}{t}$
acceleration = gradient of a velocity-time graph	
resultant force = mass × acceleration	$F = ma$
weight = mass × gravitational field strength	$W = mg$
work = force × distance	$W = Fd$
force = spring constant × extension	$F = kx$
momentum = mass × velocity	$p = mv$
force = $\dfrac{\text{change in momentum}}{\text{time}}$	$F = \dfrac{\Delta p}{t}$
u = initial velocity v = final velocity t = time a = acceleration x = displacement	$v = u + at$ $x = \dfrac{u + v}{2}\,t$
moment = force × distance	$M = Fd$

SI multipliers

Prefix	Multiplier
m	1×10^{-3}
k	1×10^{3}
M	1×10^{6}

Examiner only

Answer **all** *questions.*

1. *(a)* Light from distant galaxies is changed as it travels through the Universe.

Tick (✓) the **three** correct statements below. [3]

The wavelengths of light from distant galaxies have decreased.	☐
Distant galaxies look red.	☐
Absorption spectra from distant galaxies show red shift.	☐
The Universe has expanded since the light was given out from distant galaxies.	☐
Light received from our Sun is shifted to the blue end of the spectrum.	☐
The Sun's absorption spectrum shows no red shift.	☐

(b) Name the model of the origin of the Universe that is supported by cosmological red shift. [1]

....

4

3420U201 03

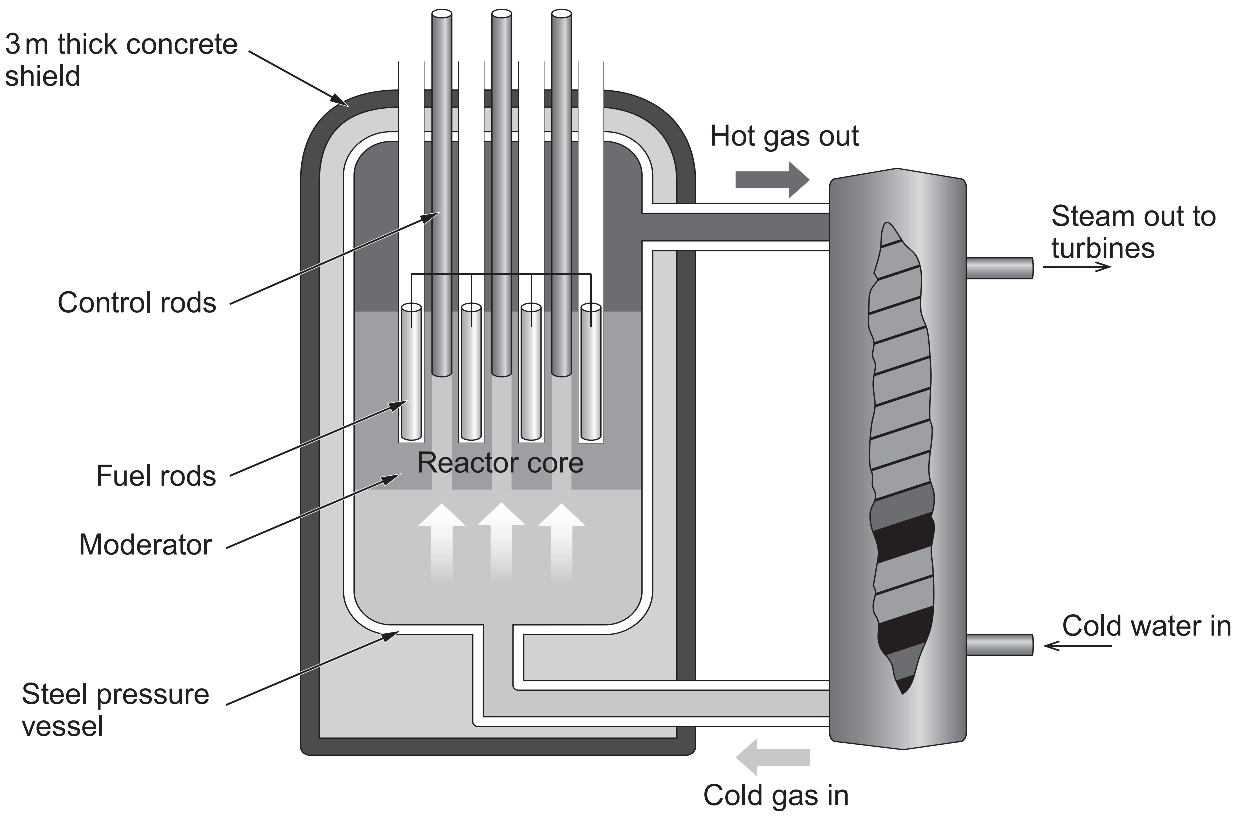
Examiner only

2. The half-life of a radioactive substance is the time taken for its activity to reduce by half.

(a) **Complete the following diagram** to show the count rate at the times shown.
The substance has a starting count rate of 800 counts per minute (cpm) and a half-life of 5 hours. [2]

count rate at the start		count rate after 5 hours		count rate after 10 hours
800 cpm	→	.... cpm	→	.... cpm

(b) The following diagram shows a nuclear fission reactor in which the uranium fuel is contained inside a steel pressure vessel which is surrounded by a 3 m thick concrete shield.



Examiner only

A uranium nucleus can split apart, forming two smaller nuclei in a nuclear reactor. One possible nuclear reaction is shown below in which a nucleus of uranium forms nuclei of tellurium (Te) and zirconium (Zr).

$$^{235}_{92}\mathrm{U} + ^{1}_{0}\mathrm{n} \longrightarrow ^{137}_{52}\mathrm{Te} + ^{97}_{40}\mathrm{Zr} + 2^{1}_{0}\mathrm{n}$$

(i) State the number of neutrons released in this reaction. .... [1]

(ii) Complete the following sentences about the nuclear reactor by underlining the correct word or phrase in each bracket. [4]

A nucleus of uranium is made to split into two smaller nuclei when it captures (**one / two / three**) neutron(s).

For fission to occur, the neutron(s) to be captured must be (**fast moving / stationary / slow moving**) and this takes place in the (**moderator / control rods / fuel rods**).

An uncontrolled chain reaction is prevented by using (**control rods / a moderator / concrete shielding**).

(c) State **two** reasons why it is difficult to store waste material from nuclear power stations. [2]

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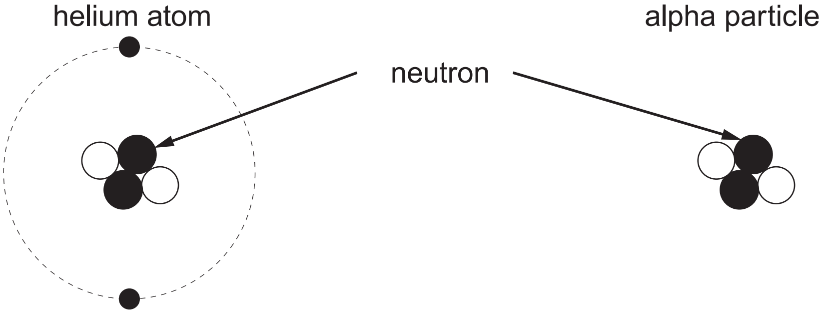
3420U201 05

9

Turn over.

Examiner only

3. A student correctly draws diagrams of a helium atom and an alpha particle.



(i) One similarity between the structure of the helium atom and the alpha particle has been given in the table below. State **one** other similarity and **one** difference between them. [2]

Similarities	Difference
They both have 2 neutrons.	....

(ii) A nucleus of polonium-210 decays by alpha particle emission to form lead (Pb), which is a stable element, according to the following reaction.

$$^{210}_{84}\text{Po} \longrightarrow {}^{4}_{2}\alpha + {}^{\cdots}_{\cdots}\text{Pb}$$

Complete the nuclear decay equation above. [2]

(iii) State why this isotope of polonium is radioactive. [1]

....

....

(iv) Explain why, in 2006, the Russian Alexander Litvinenko died within weeks of swallowing some polonium-210. [2]

....

....

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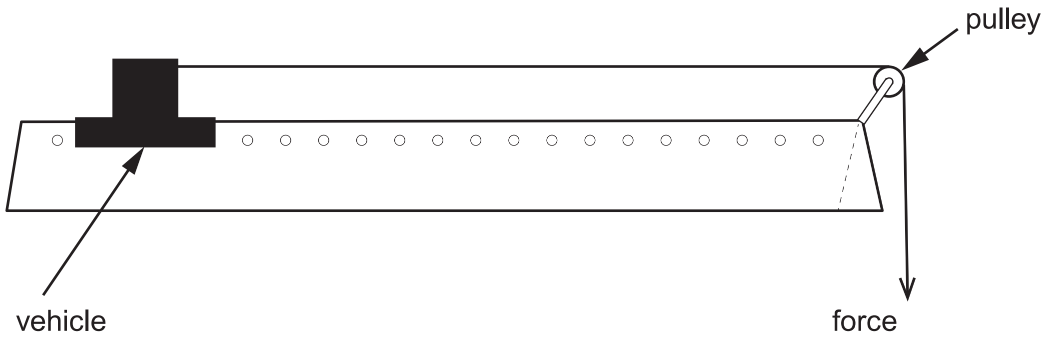
7

Examiner only

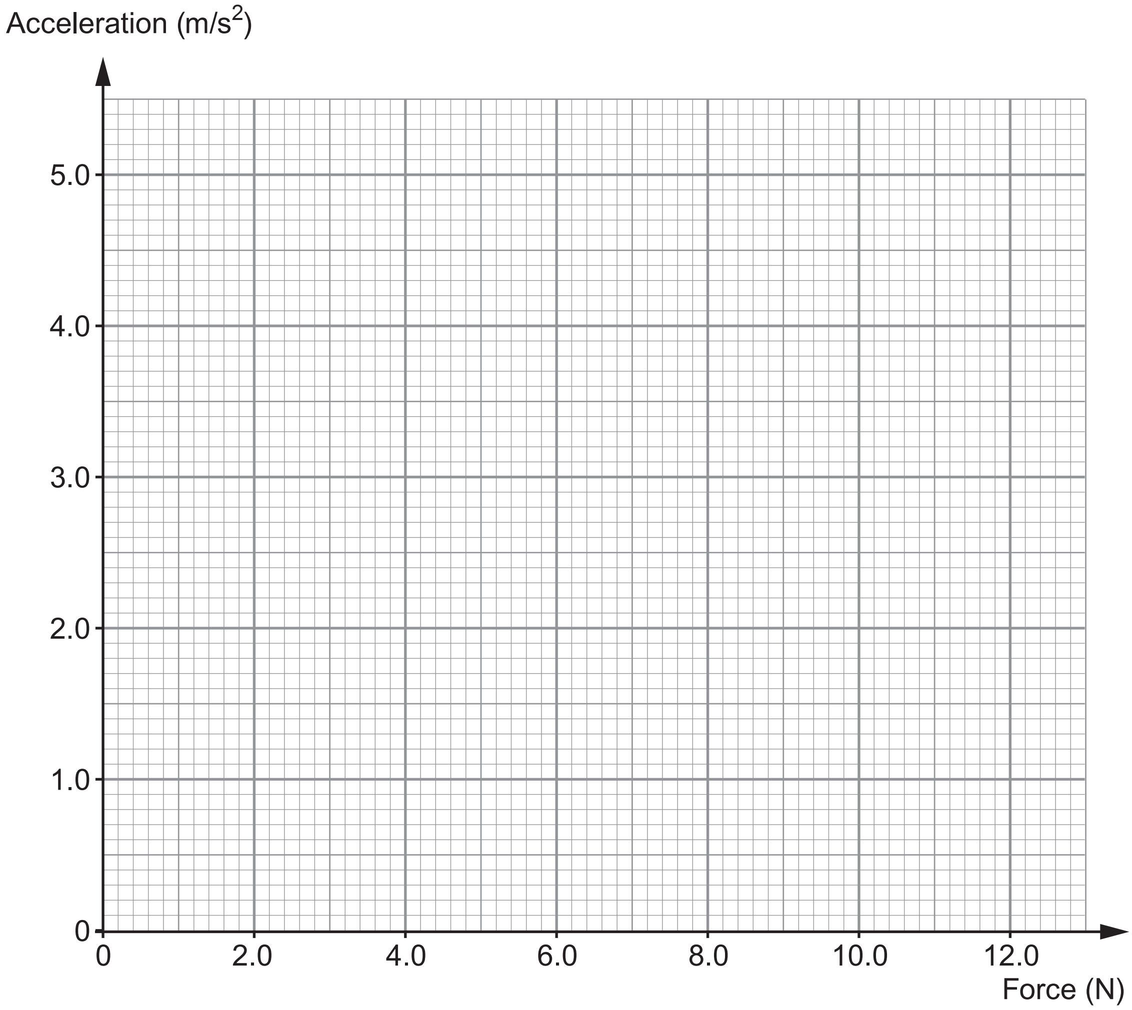
4. In a class experiment, a group of students use a force to accelerate a vehicle along a frictionless air track.



The acceleration of the vehicle is measured by a computer and the results are shown below.

Force (N)	2.0	4.0	8.0	10.0	12.0
Acceleration (m/s^2)	0.8	2.6	3.2	4.0	4.8

(a) Plot the data on the grid below and draw a suitable line. [3]



Examiner only

(b) (i) Use the graph to find the force that produces an acceleration of 2.0 m/s^2. [1]

Force = ... N

(ii) Use the equation:

$$\text{mass} = \frac{\text{resultant force}}{\text{acceleration}}$$

to calculate the mass of the vehicle. [2]

Mass = ... kg

(iii) Another group of students in the class carry out the same experiment but they use a vehicle of greater mass. **Draw a line on the graph** to show how the acceleration would change for the same values of force. [1]

(c) Eric suggests that if a force of 16 N is applied to the original vehicle the acceleration will be 5.6 m/s^2. Use the data to consider whether this suggestion is correct. [2]

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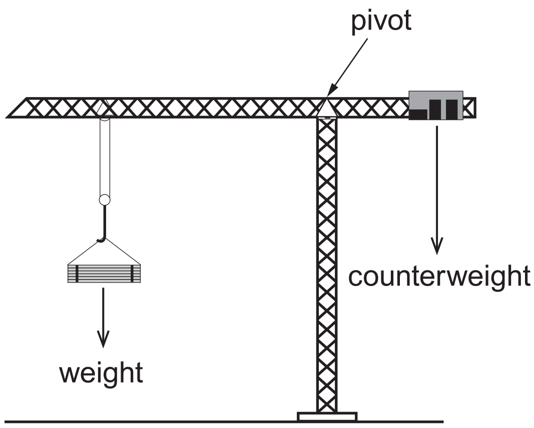
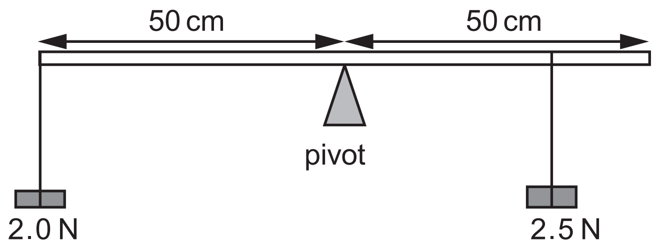
3420U201 09

9

5. The diagram shows a tower crane used on a new building in the centre of Cardiff. A weight is lifted on the long arm and a counterweight on the short arm keeps the crane from toppling over.



(a) A group of students investigate a model of the crane by hanging weights from a horizontal metre ruler.



They hang a 2.0 N weight at one end, 50 cm from the pivot and another weight of 2.5 N somewhere on the other side of the pivot to make it balance.

(i) Use the equation:

$$\text{moment} = \text{force} \times \text{distance}$$

to calculate the anticlockwise moment (in N cm) of the 2 N weight about the pivot. [2]

Moment = ... N cm

Examiner only

(ii) Use your answer to part (i) and the equation:

$$\text{distance} = \frac{\text{moment}}{\text{force}}$$

to calculate the distance that the 2.5 N weight should be placed from the pivot for the ruler to balance. [2]

Distance = ... cm

(iii) Explain why it would be impossible to balance the 2 N weight on the end of the ruler if the 2.5 N weight is replaced with a 1.5 N weight. [2]

....

....

....

(b) A particular tower crane is capable of lifting a maximum mass of 15 000 kg. Use the equation:

$$\text{weight} = \text{mass (kg)} \times \text{gravitational field strength}$$

to calculate the weight of this mass. [2]
[gravitational field strength, g = 10 N/kg]

Weight = ... N

8

3420U201
11

Examiner only

6. Cars are tested so that the manufacturers know what happens to the passengers during a collision.

In one such test, a car of mass 1 500 kg travelling with an energy of 127 500 J strikes a solid barrier and stops. The force applied to the car by the barrier is 500 000 N.

(a) (i) Name the energy that the car possesses due to its motion. [1]

....

(ii) State the work done by the barrier to stop the car. [1]

Work done = ... J

(b) (i) Use the equation:

$$\text{distance} = \frac{\text{work}}{\text{force}}$$

to calculate the distance over which the force acts. [2]

Distance = ... m

 (3420U20-1)

Examiner only

(ii) I. State **one** feature of cars that is designed to keep the driver safer in a head-on collision. [1]

....

II. Explain how this safety feature keeps the driver safer. [2]

....

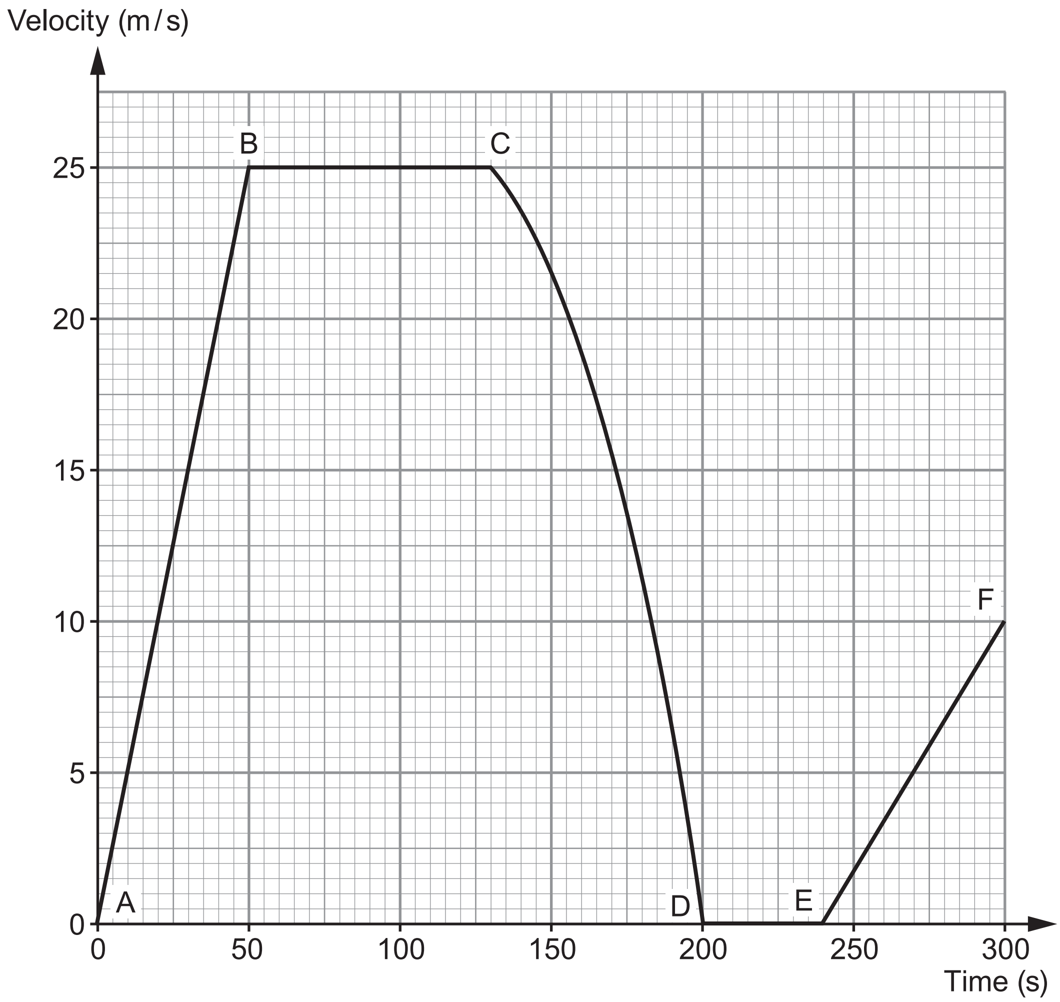
....

....

7

Examiner only

7. The velocity-time graph below shows part of the motion of an empty school bus.



(a) Use values from the graph to describe the motion of the bus over the time shown. [Note that no calculations are required as part of your answer.] [6 QER]

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Examiner only

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(b) Refer to the graph to answer the following questions. The mass of the bus is 10 000 kg.

(i) Use an equation from page 2 to calculate its change in momentum between A and B. [2]

Change in momentum = ... kg m/s

(ii) Use an equation from page 2 to calculate the resultant force that acts on the bus between A and B. [2]

Resultant force = ... N

(iii) Use the equation:

distance = speed × time

to calculate the distance travelled between B and C.

[2]

Distance = ... m

Examiner only

(iv) Another identical bus travels the same distance as in part (iii). However, it travels at a slower, constant speed. State **two** ways that **its line on the graph** would be different from the one shown. [2]

....

....

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(c) Explain, using the idea of inertia, how the acceleration of a bus filled with school children would compare with the acceleration of the empty bus. [2]

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16

Examiner only

8. A new timer, in the shape of a ball, has been produced. When it is released it starts a built-in stop clock. When the ball next hits something the timer stops. The time taken is displayed on its screen.

A teacher uses the ball timer for an experiment. He holds the ball out of his third floor classroom window and then carefully releases it. The time taken to fall is recorded in a table. The ball is dropped 5 times during the experiment.

Drop number	1	2	3	4	5
Time (s)	1.48	1.10	1.52	1.54	1.46

(a) Explain how the teacher could find out if the results are reproducible. [2]

....

....

....

(b) **Circle** the result in the table which is most likely to be anomalous. [1]

(c) Calculate the mean time for the ball timer to fall. [2]

Mean time to fall = ... s

 (3420U20-1)

Examiner only

(d) The mean impact velocity of the ball on the ground is 14.2 m/s.
Use the equation:

$$x = \frac{u + v}{2}\, t$$

to calculate the drop height of the ball. [2]

Drop height = ... m

7

Examiner only

9. In 1992 scientists first detected a planet orbiting a star outside our Solar System. Hundreds of these exoplanets have been found since. **Table 1** shows the stars that have been discovered with 6 or more exoplanets in orbit, along with information about our Sun.

Table 1

Name of star	Distance of star from Earth (l-y)	Mass of star (compared to the Sun)	Temperature of star (K)	Number of planets in orbit
Sun	0.00002	1.0	5770	8
HR 8832	21	0.8	4700	7
HD 10180	127	1.1	5910	7
Kepler 90	2500	1.1	5930	7
HD 40307	42	752.0	4980	6
HD 127	206	0.7	870	6
Kepler 20	950	0.9	5470	6
Kepler 11	2000	1.0	5680	6

(a) (i) Tick (✓) the **two** correct statements below. [2]

Four stars have the same mass. ☐

Light travelling from Kepler 90 takes the longest time to reach Earth. ☐

The range of star temperatures is 5060 K. ☐

Hotter stars have more planets in orbit around them. ☐

Star HR 8832 is double the distance of star HD 40307 away from Earth. ☐

(ii) State **two** reasons, using evidence from **Table 1**, why scientists predict that the star Kepler 11 is most likely to have a Solar System similar to ours. [2]

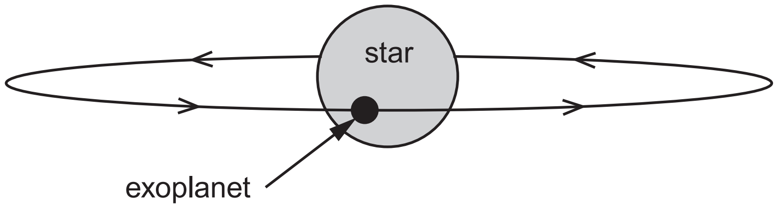
....

....

....

Examiner only

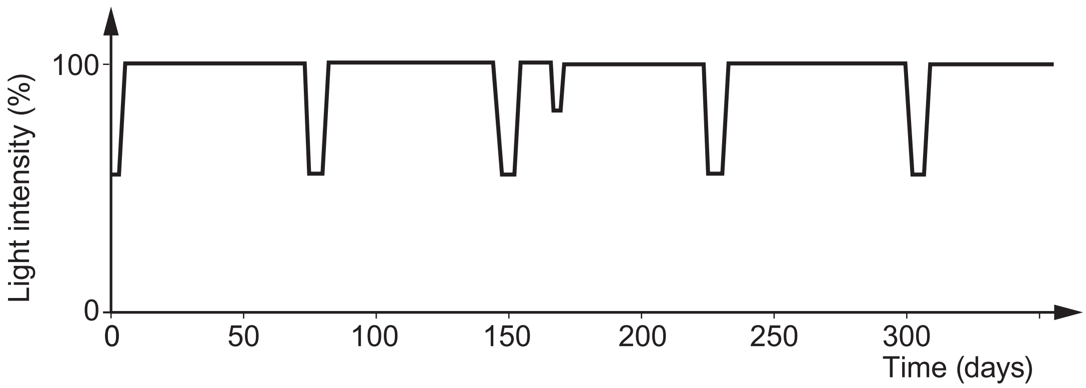
(b) Exoplanets can't be detected directly because they don't give out their own light. One method of detection is to look for a slight dip in the light intensity from the star. This happens when the exoplanet passes in front of the star during its orbit. This is called a transit.



(i) Why does the detected light intensity decrease during a transit? [1]

....

(ii) A pupil finds a graph on the internet that shows the measured light intensity detected from a star against time.



I. Calculate the orbit time, in days, of the exoplanet. [2]

Orbit time = ... days

II. The pupil correctly suggests that there may be two exoplanets in orbit around this star. What evidence is there to back up this claim? [1]

....

III. During a transit some of the light from the star passes through the exoplanet's atmosphere. An absorption spectrum is produced. Explain how the absorption spectrum arises and is used to identify gases in its atmosphere. [2]

....

....

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Examiner only

(c) **Table 2** shows data collected for all the exoplanets in orbit around the star called Kepler 11. They are listed in order of increasing distance.

Table 2

Exoplanet	Mass (compared to Earth)	Time for one orbit (days)	Temperature (K)	Orbit radius (AU)
b	1.9	10.3	871	0.09
c	2.9	13.0	807	0.11
d	7.3	22.7	659	0.16
e	8.0	32.0	596	0.20
f	2.0	46.7	525	....
g	25.0	118.4	386	0.47

(i) Compare the **trend** in temperature with orbit radius of the planets around Kepler 11 to the planets in our Solar System where Venus is the hottest. [2]

....

....

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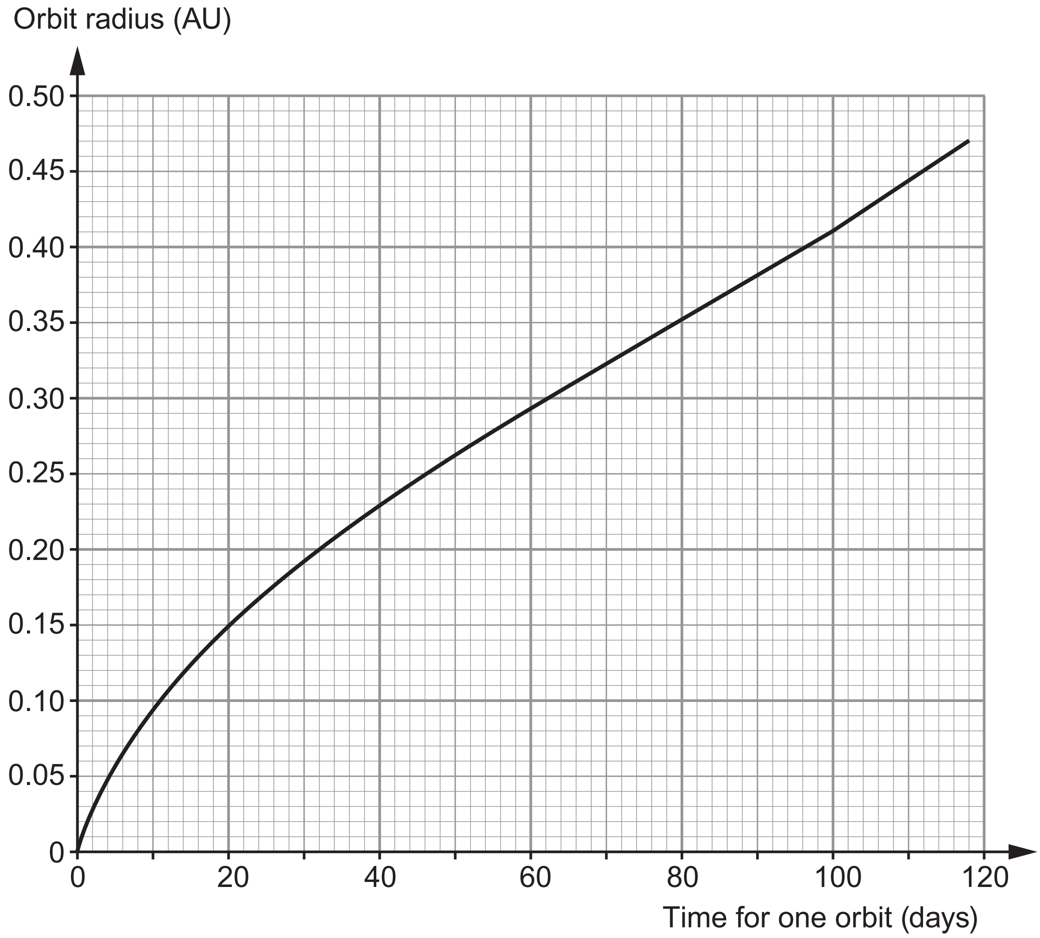
....

(ii) Some of the data from the table is used to plot a graph.



The orbit time of **exoplanet f** was found using the transit method. Its orbit time was measured as 46.7 days. Use the graph to **complete Table 2** with an estimate of a value for the missing orbit radius. [1]

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END OF PAPER

Surname	Centre Number	Candidate Number
Other Names		0

GCSE

3420U20-1

S19-3420U20-1

WEDNESDAY, 22 MAY 2019 – AFTERNOON

PHYSICS – Unit 2:
Forces, Space and Radioactivity

FOUNDATION TIER

1 hour 45 minutes

For Examiner's use only		
Question	**Maximum Mark**	**Mark Awarded**
1.	**4**	
2.	**5**	
3.	**5**	
4.	**7**	
5.	**10**	
6.	**7**	
7.	**9**	
8.	**13**	
9.	**13**	
10.	**7**	
Total	**80**	

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01

ADDITIONAL MATERIALS

In addition to this paper you will require a calculator and a ruler.

INSTRUCTIONS TO CANDIDATES

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Answer **all** questions.

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INFORMATION FOR CANDIDATES

The number of marks is given in brackets at the end of each question or part-question.

The assessment of the quality of extended response (QER) will take place in question **7**(*b*).

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Equations

speed = $\frac{\text{distance}}{\text{time}}$	
acceleration [or deceleration] = $\frac{\text{change in velocity}}{\text{time}}$	$a = \frac{\Delta v}{t}$
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moment = force × distance	$M = Fd$

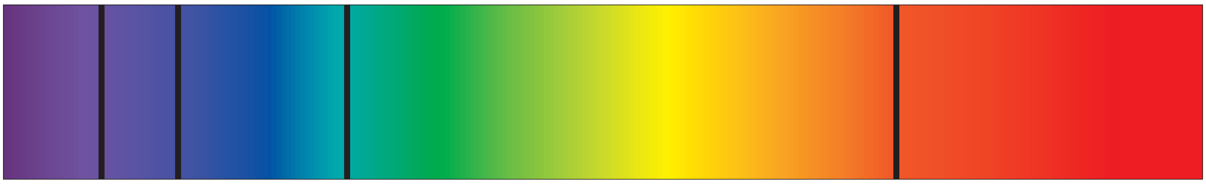
SI multipliers

Prefix	Multiplier
m	1×10^{-3}
k	1×10^{3}
M	1×10^{6}

Examiner only

Answer **all** *questions.*

1. Absorption spectra can be used to provide information about stars and galaxies. The diagram below shows a simplified absorption spectrum from our Sun.



(a) Place a tick (✓) in the box next to the **three** correct statements below. [3]

The dark lines in the spectra from distant galaxies are blue shifted. ☐

The further away a galaxy is the longer the wavelength of the dark lines. ☐

The dark lines in the spectra from distant galaxies are green shifted. ☐

The dark lines can be used to identify the elements present in the star / galaxy. ☐

The dark lines in the spectra from distant galaxies are red shifted. ☐

The further away a galaxy is the shorter the wavelength of the dark lines. ☐

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03

(b) The shift of the absorption lines in spectra from distant galaxies provides evidence for the Big Bang model of the creation of the Universe. State another piece of evidence supporting the Big Bang. [1]

....

4

Examiner only

2. A car is travelling at 10 m/s along a flat road. A driving force of 50 000 N acts on the car for a distance of 20 m causing it to accelerate.

(a) (i) Use an equation from page 2 to calculate the work done by the 50 000 N force. [2]

Work done = ... J

(ii) The car gains 600 000 J of kinetic energy as it accelerates. Calculate how much energy is transferred in other ways. [1]

Energy transfer = ... J

(b) State **two** ways that the **design** of the car could be changed to improve its efficiency. [2]

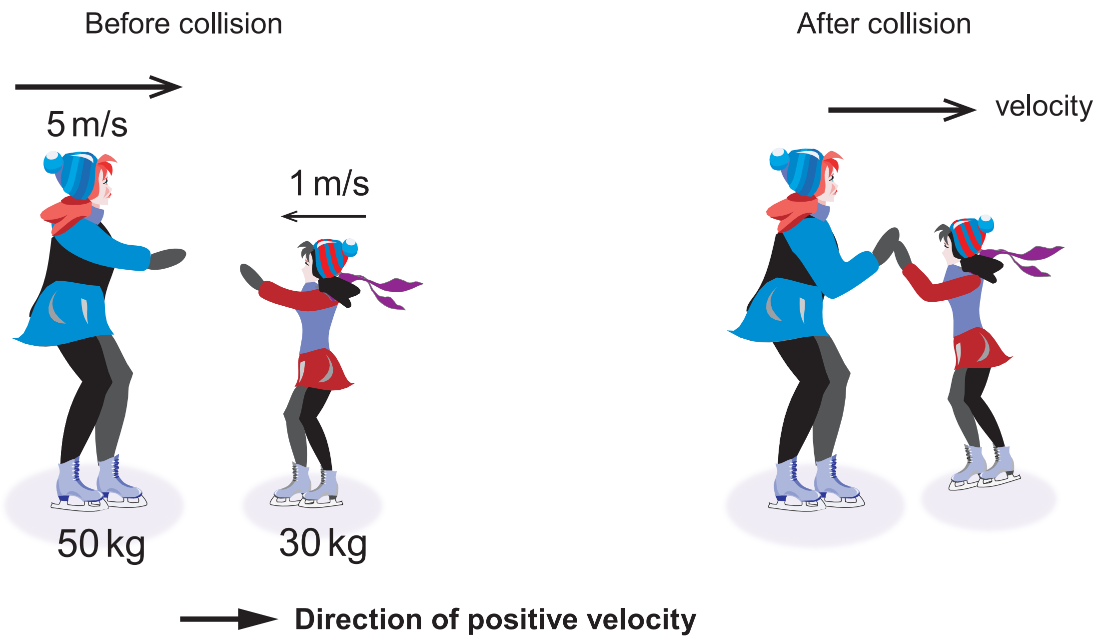
1. ....

2. ....

5

Examiner only

3. An ice skater of mass 50 kg and travelling with a velocity of 5 m/s to the right collides with a skater of mass 30 kg travelling with a velocity of 1 m/s **to the left**.



(a) Use an equation from page 2 to calculate the total momentum of the skaters **before** the collision. [3]

Total momentum = ... kg m/s

(b) After they collide the skaters move off together with a common velocity to the right. Use the equation:

$$\text{velocity} = \frac{\text{momentum}}{\text{mass}}$$

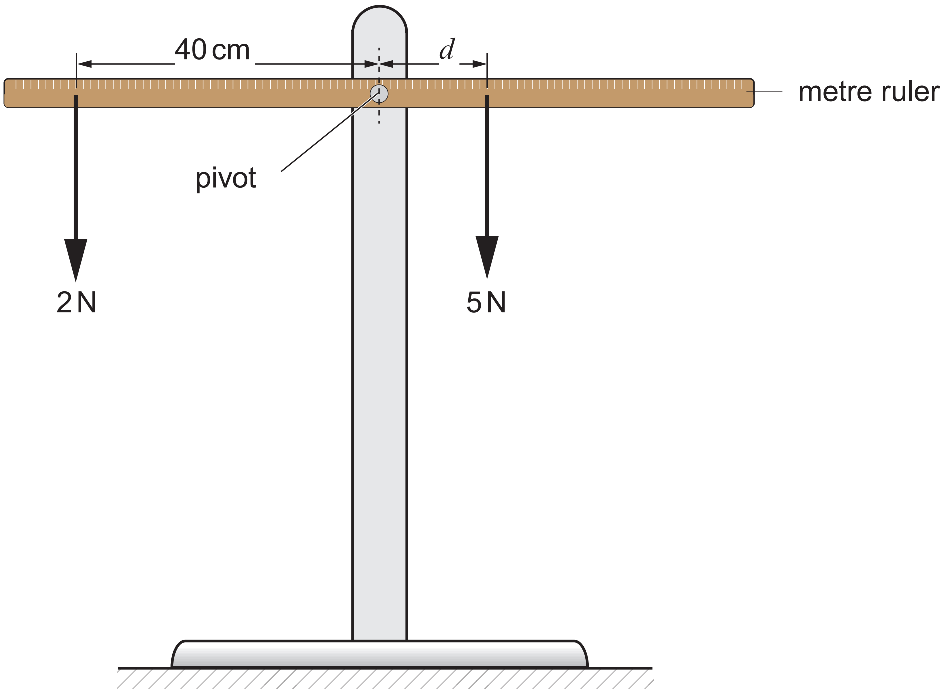
to calculate their velocity. [2]

Velocity = ... m/s

5

Examiner only

4. Some students are investigating the principle of moments. They use a metre ruler and some different weights. The ruler is pivoted at its centre. To begin with they set up their experiment as shown below. They vary the position of the 5 N weight until the ruler is balanced.



(a) (i) State the principle of moments. [1]

....

....

....

(ii) Use an equation from page 2 to calculate the moment of the 2 N weight about the pivot. Give your answer in N cm. [2]

Moment = ... N cm

Examiner only

(iii) Use the equation:

$$\text{distance} = \frac{\text{moment}}{\text{force}}$$

to calculate the distance that the 5 N weight must be from the pivot in order to balance the ruler. [2]

Distance = ... cm

(b) The students replace the 2 N weight with a 4 N weight. Jade suggests that to balance the ruler they will have to halve the distance that the 5 N weight is from the pivot. Explain whether you agree with her suggestion. [2]

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07

7

Examiner only

5. Nuclear power in the U.K. provides around one sixth of our total electricity. It is important as it provides a constant and reliable source of electricity to help supply the base load demand for the National Grid. The incomplete nuclear equation below shows one possible fission reaction of uranium, U.

$$^{235}_{92}\mathrm{U} + ^{1}_{0}\mathrm{n} \rightarrow ^{236}_{92}\mathrm{U} \rightarrow ^{96}_{37}\mathrm{Rb} + ^{\cdots\cdots}_{\cdots\cdots}\mathrm{Cs} + 3^{1}_{0}\mathrm{n}$$

(a) **Complete** the nuclear equation above. [2]

(b) (i) Name the particle that is absorbed by the uranium-235 nucleus. [1]

....

(ii) In a nuclear reactor on average two of the three particles, $^{1}_{0}\mathrm{n}$, which are produced in one fission are absorbed by the **control rods**. Explain why this allows the nuclear reactor to operate safely. [2]

....

....

....

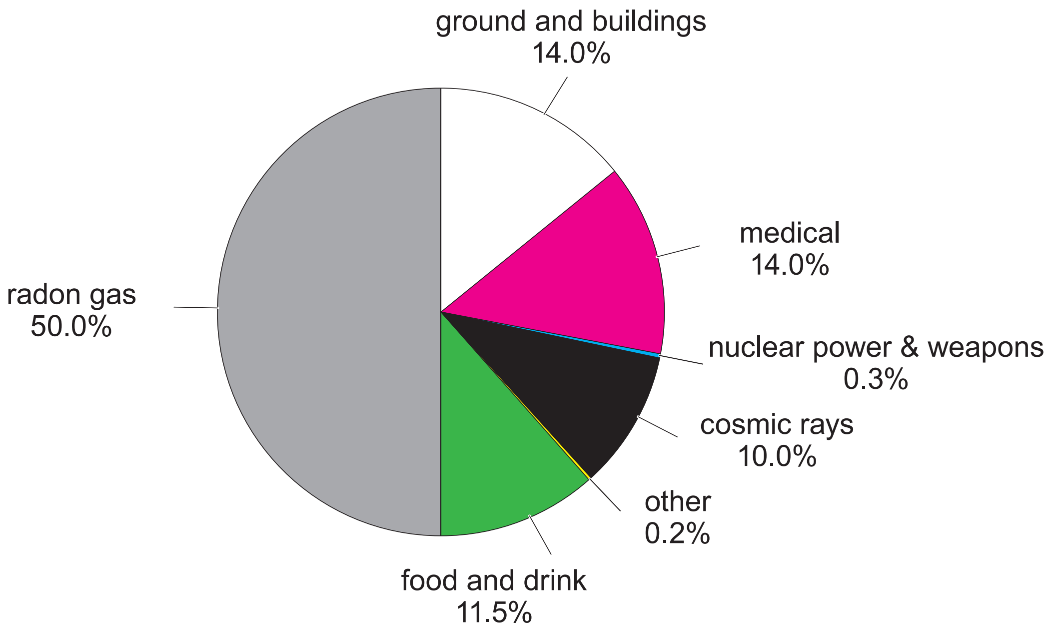
(c) The use of nuclear power is controversial and some people believe that we should not build any new nuclear power stations because of the radioactive waste that they produce. State **two** properties of nuclear waste which makes storage a problem. [2]

1. ....

2. ....

Examiner only

(d) The pie chart below shows the sources of background radiation near a nuclear power station.



(i) State how the pie chart suggests that nuclear power is a safe source of electricity. [1]

....

....

(ii) Students measure the background radiation in counts per minute. One group takes a measurement for one minute. A second group measures the counts for 10 minutes and divide the value by 10. Explain which method is better. [2]

....

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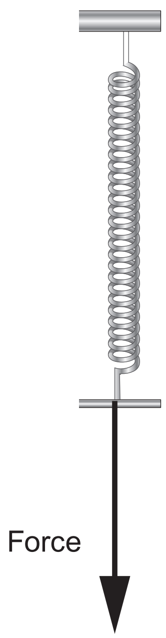
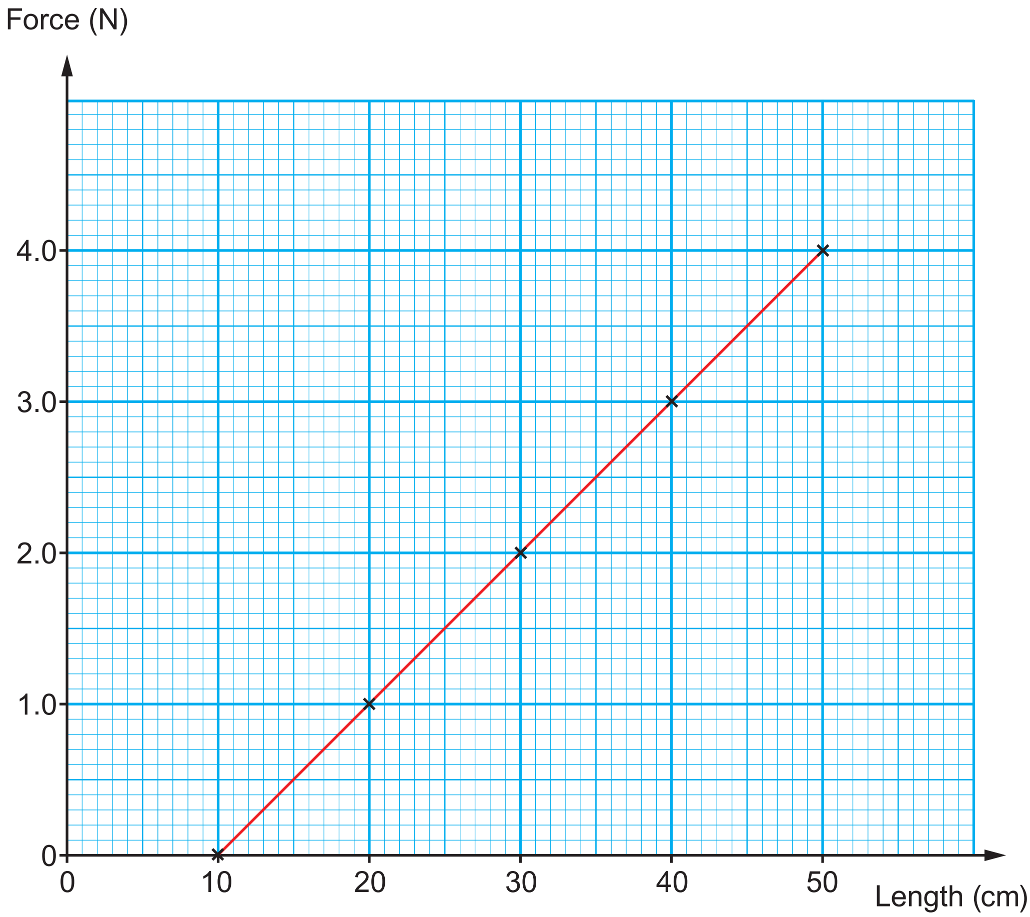
3420U201 09

10

Examiner only

6. A group of students are investigating how a spring stretches when forces are applied to it. They measure the length of the spring and plot their results on a graph.





(a) (i) Use the graph to determine the unstretched length of the spring. [1]

Unstretched length = ... cm

(ii) Use the graph and your answer to (i) to calculate the **extension** of the spring when a force of 2.5 N is applied to it. [2]

Extension = ... cm

Examiner only

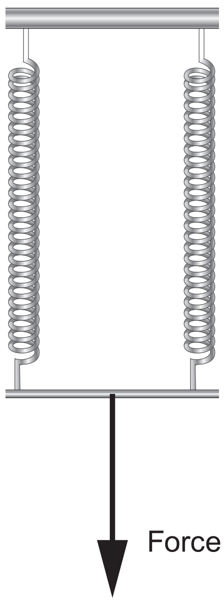
(iii) Use **your answer to (ii)** and the equation:

$$\text{spring constant} = \frac{\text{force}}{\text{extension}}$$

to calculate the spring constant in **N/m**. [2]

Spring constant = ... N/m

(b) The students add another identical spring in parallel with the first one as shown in the diagram.



They find that the spring constant is now doubled. **Add a line on the graph** to show their results. [2]

7

Examiner only

7. *(a)* Complete the following sentences below about the Sun by underlining the correct word or phrase in the brackets. [3]

The Sun produces heat and light by fusing **(hydrogen / carbon / helium)** to make **(hydrogen / carbon / helium)**. The Sun is currently stable because the gravitational force is **(less than / balanced with / greater than)** the combination of gas and radiation pressure.

(b) Compare the main stages in the life cycle of the Sun with those of a much more massive star, from birth to death. [6 QER]

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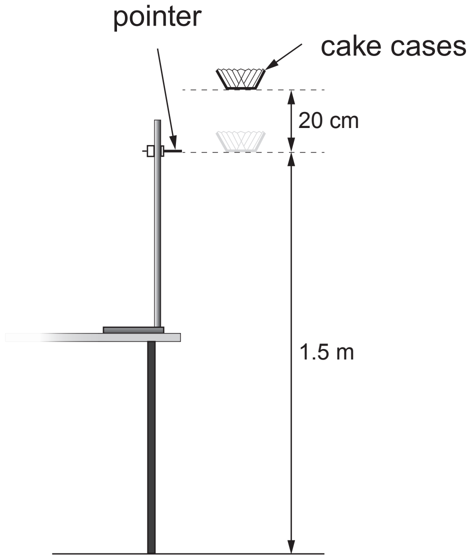
9

Examiner only

8. A class carries out an investigation into the relationship between the terminal speed of paper cake cases and their mass. They let the cake cases drop from about 20 cm above a pointer which is 1.5 m above the floor. They time how long they take to drop from the pointer to the floor. They assume that after 20 cm the cake cases will be travelling at terminal speed.



(a) Explain, in terms of two named forces, why the cake cases travel at terminal speed. [2]

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Examiner only

(b) The students take some trial readings to help them determine the number of repeat readings they need to take. Here are their results for 1 cake case.

Mass (g)	Time for cake case to travel 1.5 m (s)							
	Trial 1	Trial 2	Trial 3	Trial 4	Trial 5	Trial 6	Trial 7	Trial 8
0.5	1.16	1.19	1.17	1.29	0.72	1.22	1.24	1.15

(i) **Circle** the anomalous result. [1]

(ii) Calculate the mean time. [2]

Mean time = ... s

(iii) Use an equation from page 2 to calculate the mean speed. [1]

Mean speed = ... m/s

(c) The students then carry out their experiment. First they measure a height of 1.5 m with two metre rulers and set their pointer. They drop 1 cake case and record the time taken to drop using a stopwatch. This is repeated 5 times. They then repeat the experiment with 2, 3, 4 and 5 cake cases in a stack to vary the mass of the cake cases.

(i) State the independent variable in this experiment. [1]

....

(ii) State **one** controlled variable in this experiment. [1]

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(iii) Explain how the data could be measured more accurately. [2]

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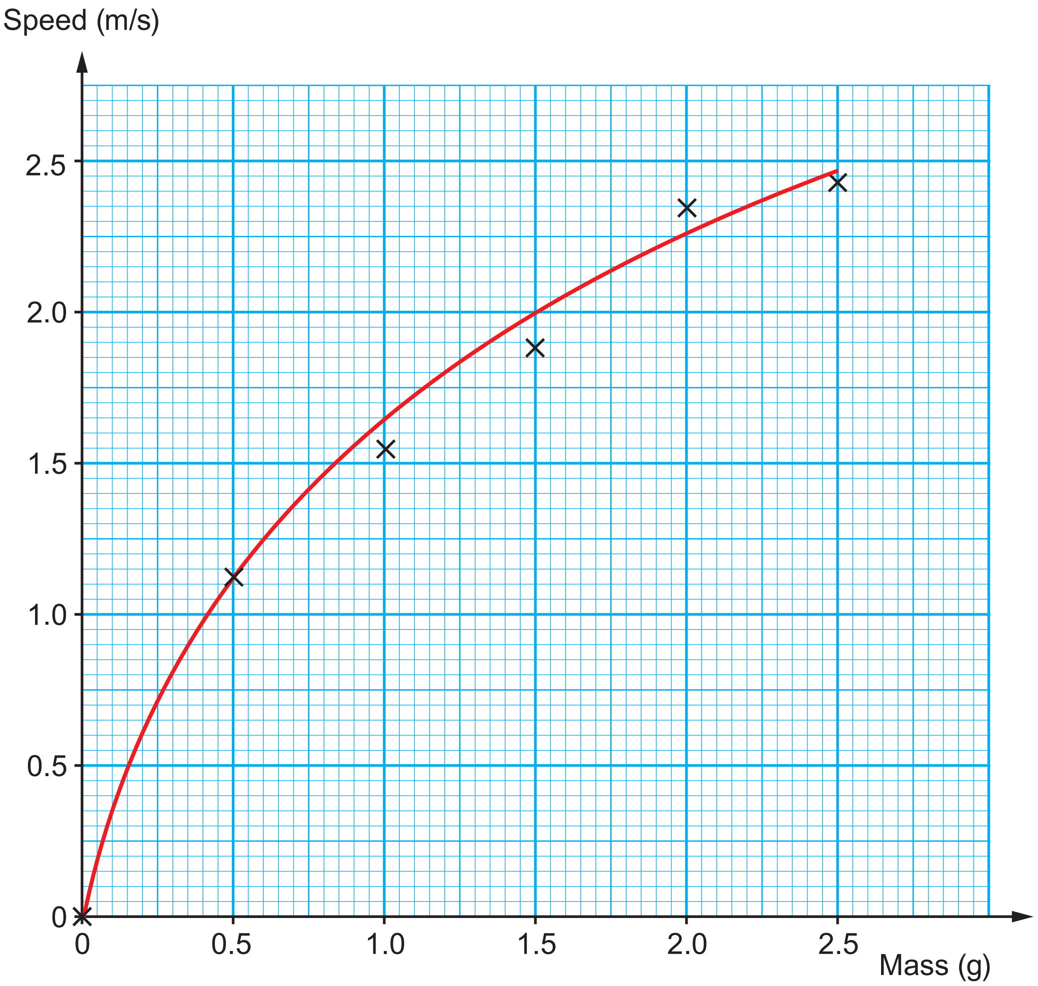
Examiner only

(d) The results from one group are shown on the graph below.



Angus concludes that when the mass doubles, the speed is always 1.5 times bigger. Explain whether the results support his conclusion. Use data from the graph to support your answer. [3]

Space for calculations.

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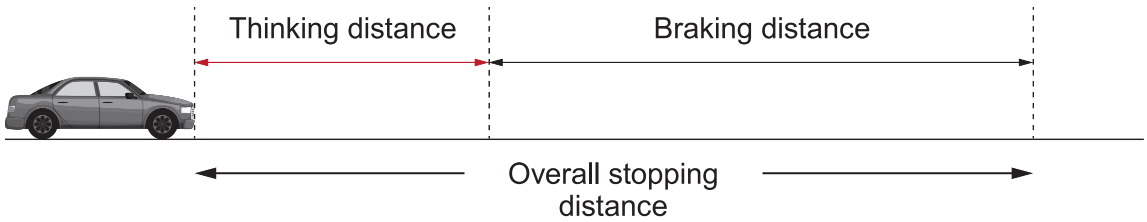
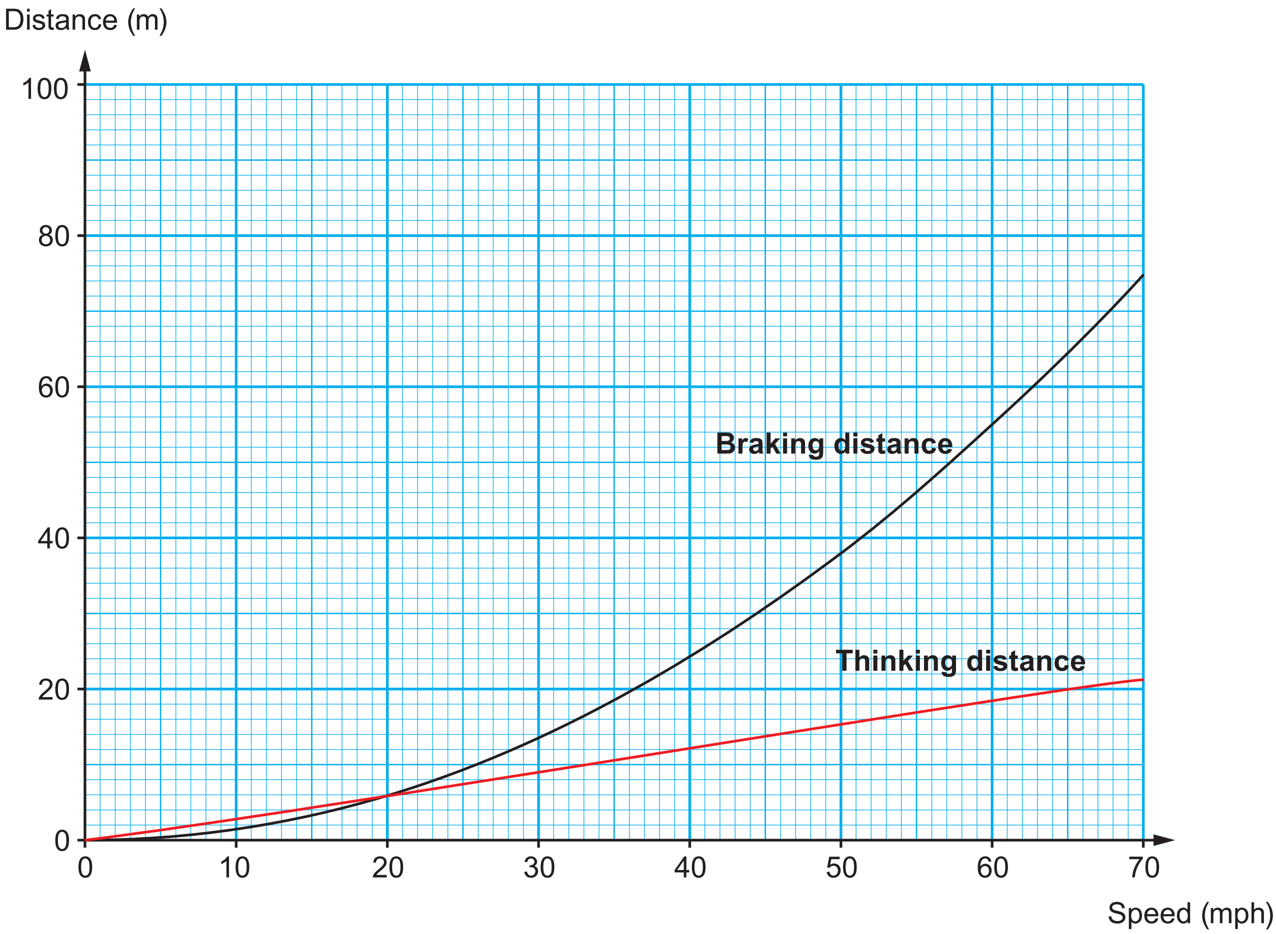
13

Examiner only

9. The overall stopping distance of a car is made up of two parts:
- the distance that the car travels when the driver is reacting (thinking distance)
- the distance that the car travels after the brakes have been applied (braking distance).



The graph below shows how the thinking distance and braking distance depend on the speed of a vehicle under good conditions.



The table below shows the conversion from mph into m/s.

Speed (mph)	20	40	60	70
Speed (m/s)	9	18	27	31

Examiner only

(a) (i) It is suggested that both thinking distance and braking distance are directly proportional to speed. Explain whether this suggestion is true. [2]

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(ii) Use information on page 16 and the equation:

$$\text{time} = \frac{\text{distance}}{\text{speed}}$$

to calculate the **thinking** time of the driver when travelling at 40 mph. [3]

Thinking time = ... s

(iii) Use the information on the graph to complete the table below. [2]

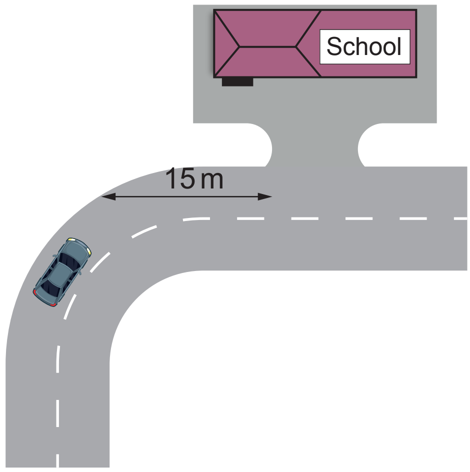
Speed (mph)	0	20	30	40	60	70
Overall stopping distance (m)	....	....	....	....	....	....

(iv) Use the data in the table to **plot** the points on the grid opposite **and draw a line** to show how the overall stopping distance depends on speed. [3]

Examiner only

(b) The speed limit along a road outside a school in Cardiff was 30 mph. The council decided to reduce this to 20 mph in 2017.

The entrance to the school is situated 15 m after a bend in the road.



Explain how the change in speed limit affects the chance of children getting knocked down as they cross the road outside the school entrance. Use data to support your answer. [3]

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13

Examiner only

10. Radiotherapy is used to treat cancer. Three types of radiotherapy are described below.

Brachytherapy is a type of internal radiotherapy. It involves putting a sealed radiation source inside the cancerous growth. The radioisotope used emits low energy gamma rays. An isotope of iodine (iodine-125) can be used to treat prostate cancer.

Unsealed source radiotherapy also uses radioactive substances to treat cancer. These are introduced into the body by injection or ingestion. Iodine-131 is injected into a patient to treat thyroid cancer.

External radiotherapy is different from the methods described above. It is given as a series of short, daily treatments in the radiotherapy department using high energy gamma rays.

Information about some isotopes of iodine is given below.

Iodine-123 has a half-life of 13 hours and emits gamma.
Iodine-125 has a half-life of 59 days and emits gamma.
Iodine-128 has a half-life of 25 minutes and emits beta.
Iodine-129 has a half-life of 15 000 000 years and emits beta and gamma.
Iodine-131 has a half-life of 8 days and emits beta and gamma.

(a) Explain why iodine-123 is unsuitable to treat prostate cancer. [2]

....

....

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(b) Explain why iodine-131 is more suitable to treat thyroid cancer than iodine-128. [2]

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(c) Patients are told that, after treatment with iodine-131, small amounts of radiation from their body may trigger radiation monitors until the activity has dropped to one thousandth $\left(\frac{1}{1000}\right)$ of its initial value. The patients are told this will occur 80 days after treatment.
Explain with the aid of a calculation whether 80 days is long enough. [3]

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7

END OF PAPER

Surname	Centre Number	Candidate Number
First name(s)		0

GCSE – CONTINGENCY

3420U40-1

Z22-3420U40-1

THURSDAY, 23 JUNE 2022 – AFTERNOON

PHYSICS – Unit 2:
Forces, Space and Radioactivity

FOUNDATION TIER

1 hour 45 minutes

For Examiner's use only		
Question	Maximum Mark	Mark Awarded
1.	3	
2.	3	
3.	9	
4.	3	
5.	10	
6.	7	
7.	7	
8.	11	
9.	7	
10.	10	
11.	10	
Total	80	

ADDITIONAL MATERIALS

In addition to this paper you will require a calculator and a ruler.

INSTRUCTIONS TO CANDIDATES

Use black ink or black ball-point pen. Do not use gel pen or correction fluid. You may use a pencil for graphs and diagrams only.

Write your name, centre number and candidate number in the spaces at the top of this page.

Answer **all** questions.

Write your answers in the spaces provided in this booklet. If you run out of space use the additional page at the back of the booklet, taking care to number the question(s) correctly.

INFORMATION FOR CANDIDATES

The number of marks is given in brackets at the end of each question or part-question.

The assessment of the quality of extended response (QER) will take place in question **8(a)**.

3420U401 01

 VP/DJ*(Y21-3420U40-1)

Equations

speed = $\dfrac{\text{distance}}{\text{time}}$	
acceleration [or deceleration] = $\dfrac{\text{change in velocity}}{\text{time}}$	$a = \dfrac{\Delta v}{t}$
acceleration = gradient of a velocity-time graph	
resultant force = mass × acceleration	$F = ma$
weight = mass × gravitational field strength	$W = mg$
work = force × distance	$W = Fd$
force = spring constant × extension	$F = kx$
momentum = mass × velocity	$p = mv$
force = $\dfrac{\text{change in momentum}}{\text{time}}$	$F = \dfrac{\Delta p}{t}$
u = initial velocity v = final velocity t = time a = acceleration x = displacement	$v = u + at$ $x = \dfrac{u + v}{2}\, t$
moment = force × distance	$M = Fd$

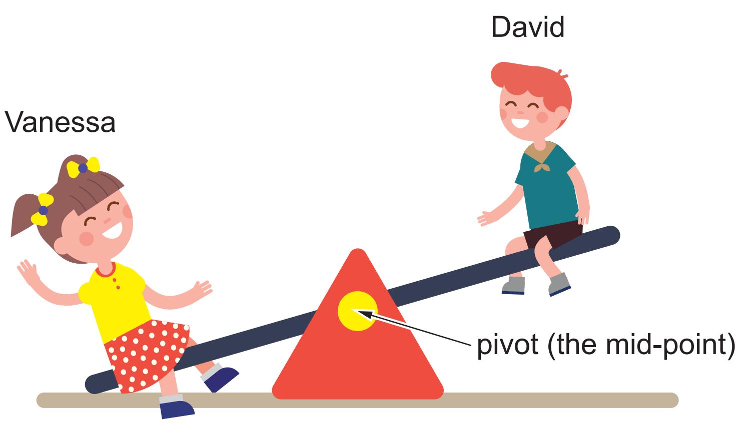
SI multipliers

Prefix	Multiplier
m	1×10^{-3}
k	1×10^{3}
M	1×10^{6}

Examiner only

Answer **all** questions.

1. (a) The picture shows Vanessa and David sitting on a see-saw.



The see-saw in the diagram is not balanced because Vanessa is heavier than David.

Place a **tick (✓)** in **one** box alongside the correct statement below. [1]

Vanessa and David can't balance the see-saw because they have different weights.	☐
If Vanessa moves nearer the pivot, the see-saw can be balanced.	☐
For the see-saw to balance, they **must** sit at the ends.	☐

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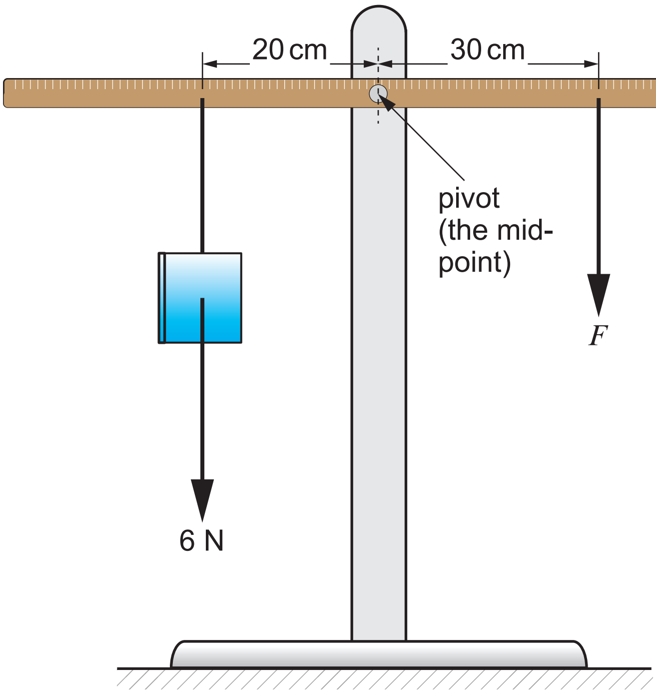
03

Examiner only

(b) A metre ruler is hung from its mid-point. A book weighing 6 N is 20 cm from the pivot. It is balanced by a force, F as shown in the diagram.



Use the equation:

$$F = \frac{6 \times 20}{\text{distance}}$$

to calculate the value of the force, F. [2]

force, F = ... N

3

Examiner only

2. Underline the correct word or phrase from each bracket to complete the sentences below. [3]

Cosmic **(microwave / material / metre)** background radiation is often referred to as CMBR.

As this radiation travelled through space its wavelength **(decreased / stayed the same / increased)**.

The background radiation in space began with **(the formation of the Sun / the Big Bang / the formation of the Milky Way)**.

3

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Examiner only

3. The following table contains data about four of the planets that orbit the Sun.

	Distance from Sun (million km)	Surface temperature (°C)	Diameter (units)	Mass (units)	Gravitational field strength, g (N/kg)
Earth	150	15	10	60	10.0
Mars	230	–65	5	6	3.7
Saturn	1435	–140	100	5700	9.0
Venus	108	464	10	49	9.0

(adapted from planetary fact sheet NASA)

Use only information from the table to answer the following questions.

(a) Underline the correct word in each of the sentences below. [3]

(i) The planet nearest the Sun is **(Earth / Mars / Saturn / Venus)**.

(ii) The coldest planet is **(Earth / Mars / Saturn / Venus)**.

(iii) The biggest planet is **(Earth / Mars / Saturn / Venus)**.

(b) (i) State the gravitational field strength on Venus. [1]

.... N/kg

(ii) Use the equation:

weight = mass × gravitational field strength on Venus

to calculate the weight of an object of mass 80 kg on Venus. [2]

weight = ... N

(iii) Name a planet on which an object of mass 80 kg would have the same weight as on Venus. [1]

....

Examiner only

(c) One student, Ceri, says that planets with the same diameter have the same mass. Explain whether you agree with this statement. [2]

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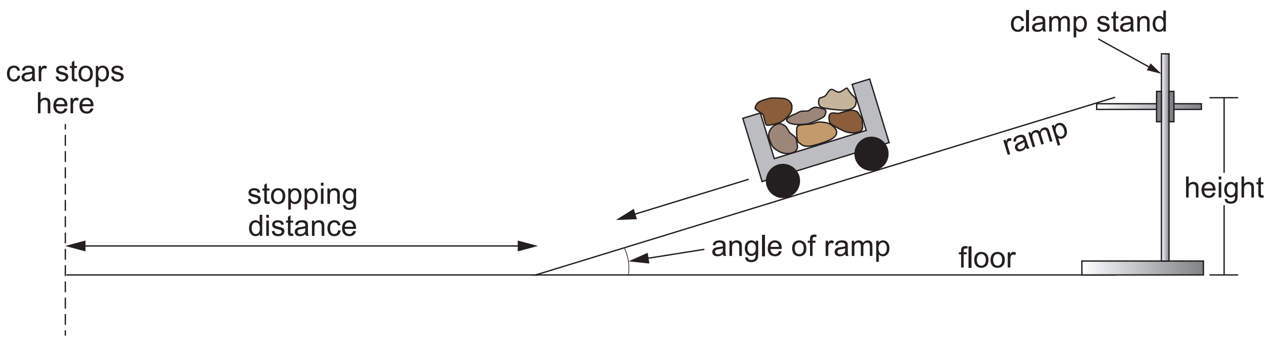
9

3420U401
07

Examiner only

4. In a class experiment, some students investigate the stopping distance of a toy car after it travels down a ramp.
In their experiment, they add stones to the toy to investigate whether its total weight affects its stopping distance along the flat floor.



(a) State the **dependent** variable. [1]

....

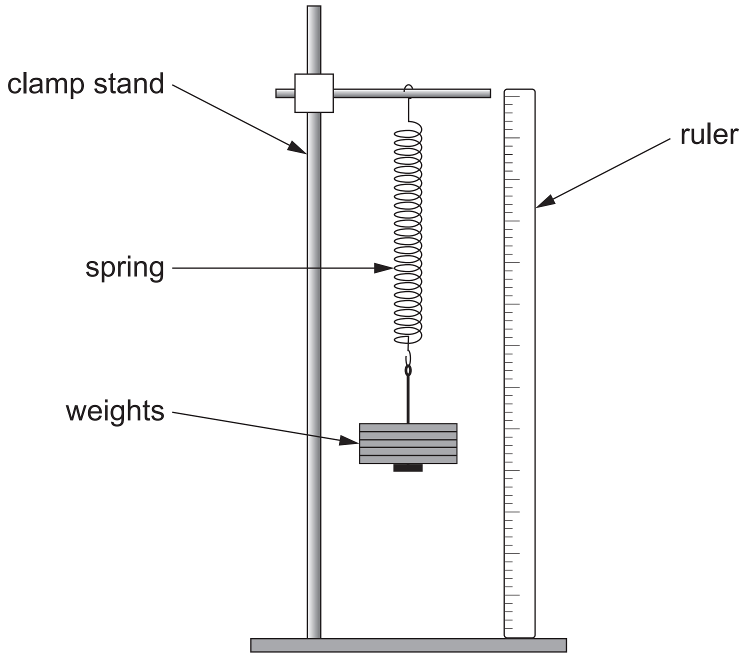
(b) The students are asked to investigate other variables that would affect its stopping distance.
Using the same apparatus given, state **two** other independent variables they could investigate. [2]

1. ....

2. ....

3

5. The diagram shows the apparatus used by a group of students when investigating the stretching of a spring.



The spring is loaded and then unloaded.
Its extension is determined when different weights are added.
Their results are shown below.

Force (N)	Extension when loading (cm)	Extension when unloading (cm)	Mean extension (cm)
0.0	0.0	0.0	0.0
1.0	3.9	4.1	4.0
2.0	8.0	8.0	8.0
4.0	16.0	16.0	16.0
5.0	20.2	19.8	20.0
6.0	23.7	24.3	24.0
7.0	28.0	28.0	28.0

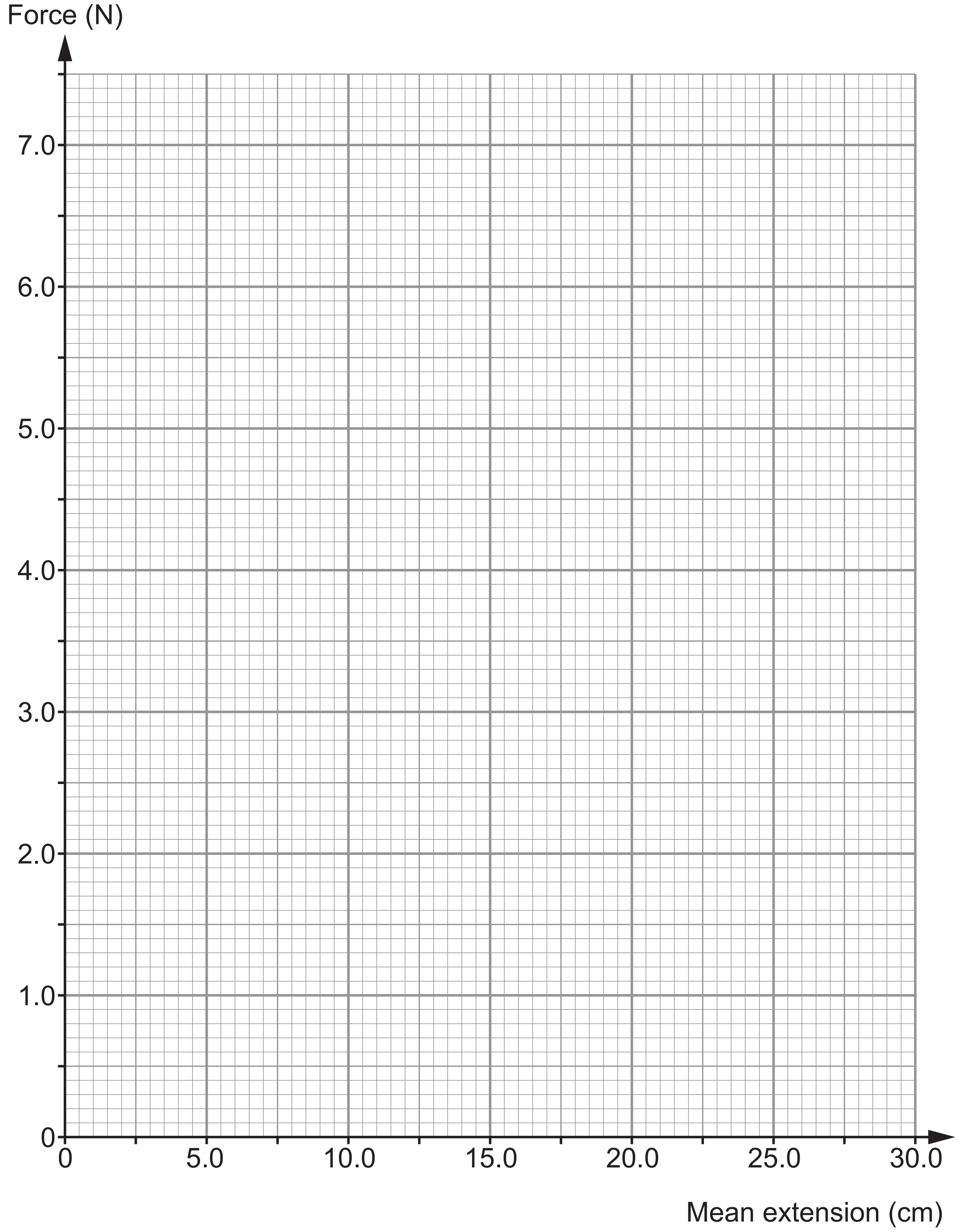
(a) State the **two** measurements that need to be taken to determine the extension of the spring. [2]

1. ....

2. ....

Examiner only

(b) Plot the data on the grid below and draw a suitable straight line. [3]



(c) Use your graph to determine the extension for a force of 2.5 N. [1]

extension = ... cm

Examiner only

(d) Use your answer from part (c) and the equation:

$$\text{spring constant} = \frac{\text{force}}{\text{extension}}$$

to calculate the spring constant for a force of 2.5 N. [2]

spring constant = ... N/cm

(e) Explain how you could check whether the results of this investigation are reproducible. [2]

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10

Examiner only

6. You are exposed to natural sources of background radiation all the time.
The level of background radiation varies depending on where you live.

(a) (i) Name **one** cause of background radiation. [1]

....

(ii) Give a reason why the level of background radiation depends on where you live. [1]

....

....

(b) Having an X-ray taken of part of the body exposes the person to ionising radiation.
The following table gives the effect on the body from X-rays.

X-ray	Radiation's effect on the body	
	Radiation dose (units)	Equivalent number of hours of background radiation
chest	100	200
hand	1	2
dental	5	10

(i) State which X-ray gives the most risk to the patient and give a reason for your answer. [2]

X-ray ....

Reason ....

....

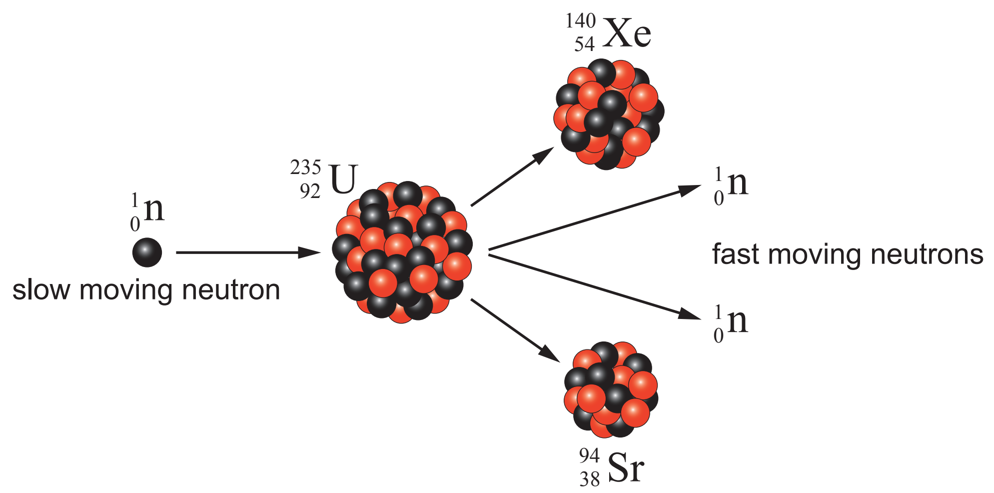
(ii) A person has 1 chest X-ray and 4 dental X-rays in his lifetime.
Toni suggests that this is equivalent to 240 hours of extra background radiation.
Use data from the table to decide whether you agree with this suggestion. [3]
Space for calculation.

....

7

Examiner only

7. The diagram shows the fission of a uranium (U) nucleus by a slow-moving neutron ($^{1}_{0}n$).
This reaction can take place in a nuclear reactor.
The fission produces nuclei of xenon (Xe) and strontium (Sr) and further neutrons.



(a) Explain the function of the moderator in a nuclear reactor. [2]

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(b) **Complete the nuclear equation** for the reaction shown in the diagram above. [3]

$$^{1}_{0}n + ^{235}_{92}U \longrightarrow \text{......................} + \text{......................} + \text{..........}\ ^{1}_{0}n$$

(c) In the nuclear reaction shown above, only one of the neutrons causes further fission.

Underline the correct word from each bracket below to complete the sentences correctly. [2]

The other neutron is **(emitted / absorbed / reflected)**.
This is done by **(control / fuel / moderator)** rods.

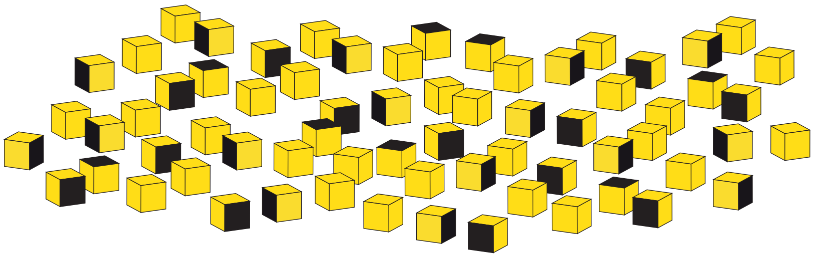
7

 (3420U40-1)

Examiner only

8. The atoms of radioactive substances decay at random.

(a) Students are given 200 cubes each with one side coloured black.
Describe how they would use the cubes to model radioactive decay **and** find the half-life. [6 QER]



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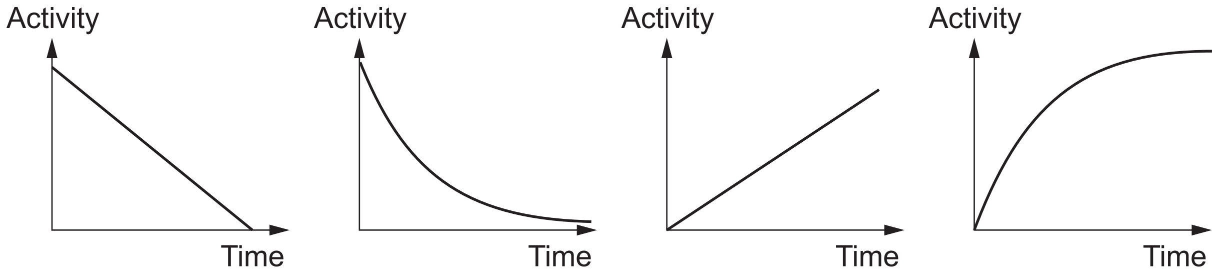
Examiner only

(b) An activity-time graph can be drawn for a radioactive substance.

Place a **tick (✓)** in the box beneath the graph that shows how the activity of a radioactive substance changes with time. [1]



☐ ☐ ☐ ☐

(c) In an extension to their experiment, each group of students in a class is given 270 6-sided dice.
After each throw they remove the dice that show a ONE or a SIX facing upwards.

Their results are shown below.

Number of throws	Number of ONES removed	Number of SIXES removed	Number of dice remaining
0			270
1	40	....	185
2	37	30	118
3	18	17	....
4	14	14	55
5	10	9	36

(i) After the **first throw**, how many ONES would you have **expected** to be removed? [1]

number of ONES expected ....

(ii) **Complete the table.** [2]

(iii) Why does combining results from several groups of students better model radioactive decay? [1]

....

11

Examiner only

9. A student aims to improve his fitness.
Starting from rest, he can accelerate at $a = 2\,\text{m/s}^2$ for a time, $t = 4\,\text{s}$.

Use the information above to answer the following questions.

(a) (i) State the value of the initial velocity, u. [1]

initial velocity, u = ... m/s

(ii) Use the equation:

$$v = u + at$$

to calculate his final velocity, v. [2]

final velocity, v = ... m/s

(b) Use the equation:

$$x = \frac{(u + v)}{2}\,t$$

to calculate the distance travelled, x, in this time of 4 s. [2]

distance, x = ... m

(c) The student thinks that there are two ways that he could increase his acceleration.
1. Reducing body mass.
2. Increasing the strength of his legs.

Explain whether you agree with his ideas. Use $a = \dfrac{F}{m}$. [2]

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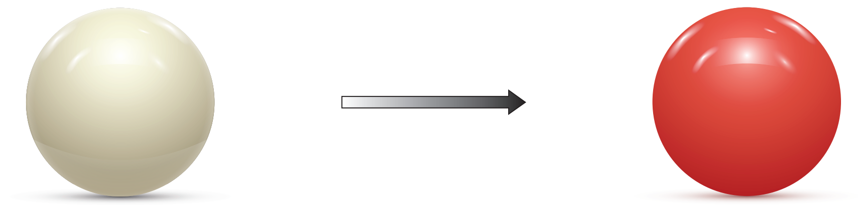
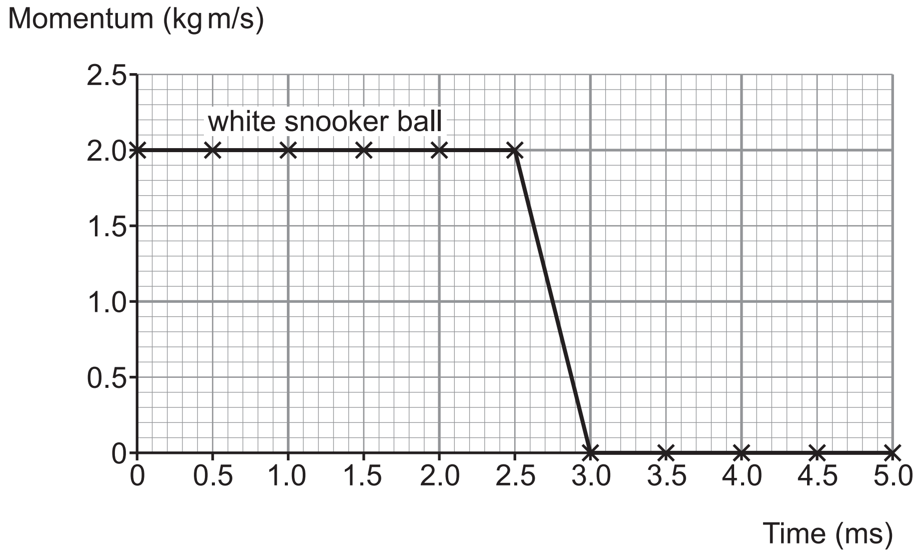
7

Examiner only

10. A white snooker ball, rolling to the right, collided with a **stationary** red snooker ball at a time of 2.5 ms.
Each snooker ball has a mass 0.16 kg.



The graph below shows the momentum of the **white snooker ball** before and after the collision.



(a) Use information from the graph to answer the following questions.

(i) State the initial momentum of the white snooker ball. [1]

initial momentum = ... kg m/s

 (3420U40-1)

Examiner only

(ii) Use the equation:

$$\text{initial velocity} = \frac{\text{initial momentum}}{\text{mass}}$$

to calculate the initial velocity of the white snooker ball. [2]

initial velocity = ... m/s

(iii) The collision takes a time of 0.5 ms. State this time in seconds. [1]

time = ... s

(iv) Use the equation:

$$\text{resultant force} = \frac{\text{change in momentum}}{\text{time}}$$

to calculate the resultant force on the white snooker ball during the collision. [2]

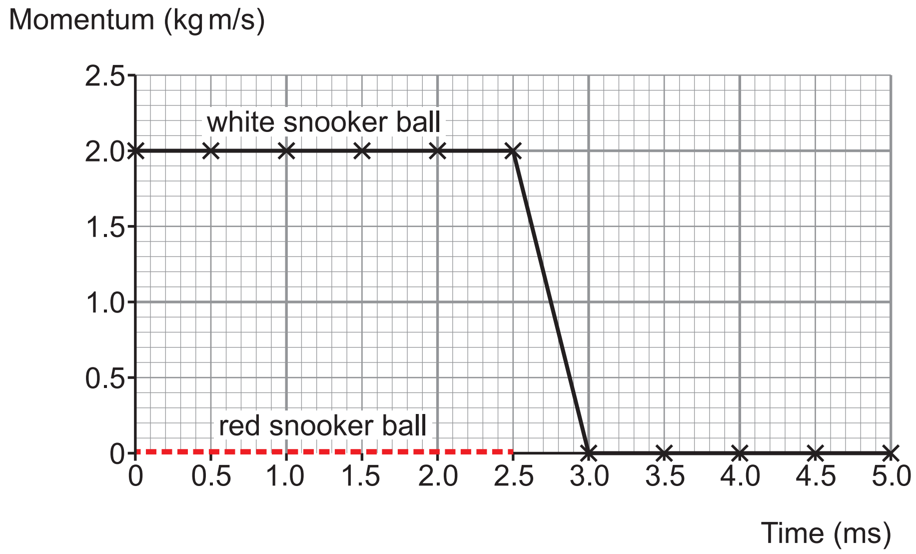
resultant force = ... N

(b) (i) Underline the correct word or phrase from each of the brackets below which correctly completes the sentence. [2]

The law of conservation of momentum states that the total momentum before a collision is **(less than / equal to / greater than)** the total momentum after a collision provided **(no / small / large)** external forces act.

Examiner only

(ii) The momentum of the red snooker ball, between 0.0 and 2.5 ms, has been added to the original graph. It is shown as a red dotted line.
Complete the graph below to show the momentum of the red snooker ball from 2.5 ms to 5.0 ms. [2]



10

Examiner only

11. A group of students investigate how the surface area of a falling paper cake case affects its terminal speed.

- Cake case 1 has a mass of 0.5 g and a surface area of 100 cm^2.
- Cake case 1 is dropped from a height of 1.80 m but only timed over the final 1.50 m of the fall.

The students' results are shown in the table below.

Drop time (s)			Mean drop time (s)	Drop distance (m)
Attempt 1	Attempt 2	Attempt 3		
0.96	0.92	0.94	....	1.50

(a) (i) The students decide there are no anomalies. Explain why. [1]

....

(ii) **Complete the table** to show the mean drop time. [1]
Space for calculation.

(b) The experiment is repeated with cake case 2.
It has the same shape and the same mass as cake case 1.
However, cake case 2 has a surface area of 50 cm^2.
The students correctly calculate the terminal speed for both cake cases.

Cake case 1		
Mass (g)	Surface area (cm^2)	Terminal speed (m/s)
0.5	100	1.6

Cake case 2		
Mass (g)	Surface area (cm^2)	Terminal speed (m/s)
0.5	50	2.3

(i) A cake case reaches terminal speed when its weight is balanced by air resistance. **Tick (✓)** the **three** correct statements. [3]

Cake case 2 has the same terminal speed as cake case 1.	☐
Cake cases 1 and 2 have identical weight.	☐
At terminal speed, cake case 1 experiences a greater value of air resistance than cake case 2.	☐
At terminal speed, both cake cases experience identical values of air resistance.	☐
At terminal speed, cake case 1 experiences a smaller value of air resistance than cake case 2.	☐
At terminal speed, both cake cases have zero acceleration.	☐

(ii) Before the experiment was carried out the students made the following prediction:

"If the surface area of the cake case is halved its terminal speed will double."

Use data from the tables on the previous page to explain whether their prediction was correct. [2]

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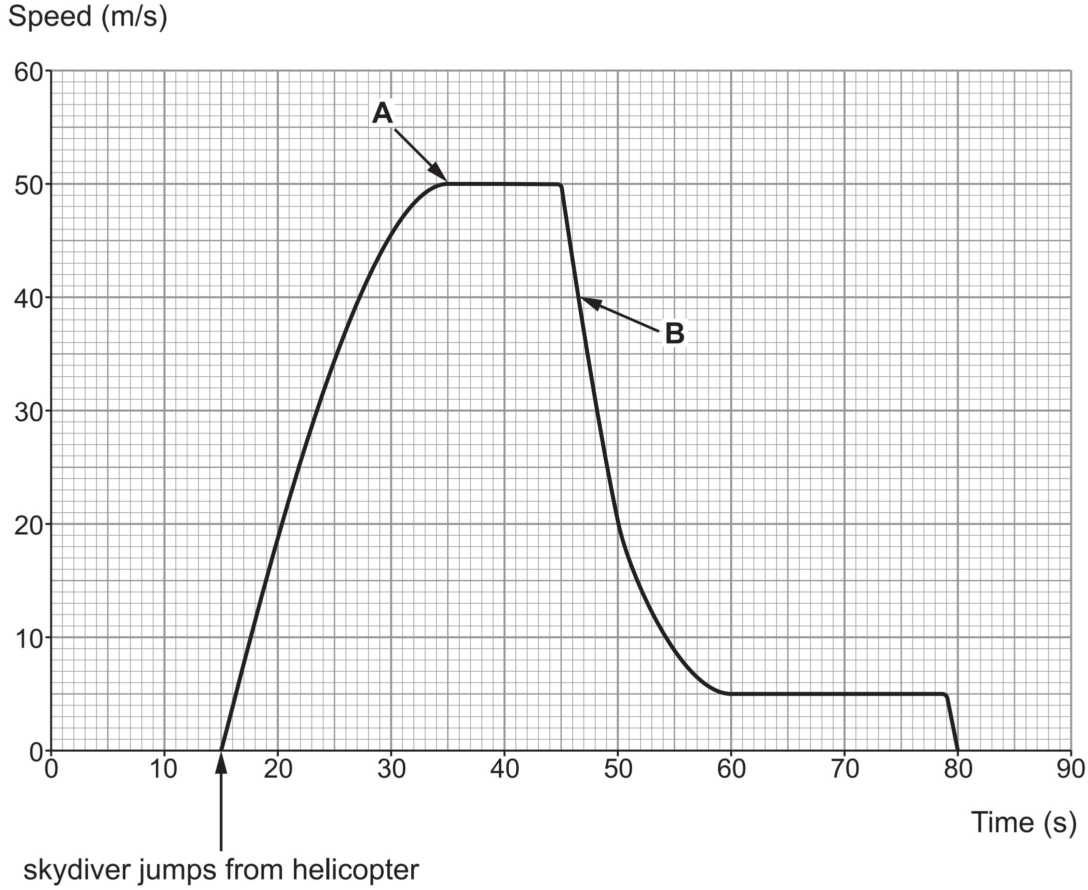
Examiner only

(c) A skydiver sits at the doorway of a helicopter for 15 s before jumping. His speed is recorded and displayed on the graph below.



Tick (✓) the **three** correct statements. [3]

The skydiver lands on the ground 80 s after jumping from the helicopter.	☐
The terminal speed after the parachute is opened is $\frac{1}{10}$th of the terminal speed before the parachute is opened.	☐
The skydiver's weight is greatest at the point labelled **B** on the graph.	☐
The parachute is opened 30 s after the skydiver leaves the helicopter.	☐
At point **B** the weight of the skydiver is greater than the air resistance.	☐
At point **A** the skydiver stops accelerating.	☐

10

END OF PAPER

Surname	Centre Number	Candidate Number
First name(s)		0

GCSE

3420U20-1

Z22-3420U20-1

WEDNESDAY, 8 JUNE 2022 – AFTERNOON

PHYSICS – Unit 2:
Forces, Space and Radioactivity

FOUNDATION TIER

1 hour 45 minutes

For Examiner's use only		
Question	**Maximum Mark**	**Mark Awarded**
1.	**6**	
2.	**9**	
3.	**4**	
4.	**5**	
5.	**11**	
6.	**7**	
7.	**12**	
8.	**6**	
9.	**14**	
10.	**6**	
Total	**80**	

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ADDITIONAL MATERIALS

In addition to this paper you will require a calculator and a ruler.

INSTRUCTIONS TO CANDIDATES

Use black ink or black ball-point pen. Do not use gel pen or correction fluid.

You may use a pencil for graphs and diagrams only.

Write your name, centre number and candidate number in the spaces at the top of this page.

Answer **all** questions.

Write your answers in the spaces provided in this booklet. If you run out of space use the additional page at the back of the booklet, taking care to number the question(s) correctly.

INFORMATION FOR CANDIDATES

The number of marks is given in brackets at the end of each question or part-question.

The assessment of the quality of extended response (QER) will take place in question **7(b)(i)**.

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Equations

speed = $\frac{\text{distance}}{\text{time}}$	
acceleration [or deceleration] = $\frac{\text{change in velocity}}{\text{time}}$	$a = \frac{\Delta v}{t}$
acceleration = gradient of a velocity-time graph	
resultant force = mass × acceleration	$F = ma$
weight = mass × gravitational field strength	$W = mg$
work = force × distance	$W = Fd$
force = spring constant × extension	$F = kx$
momentum = mass × velocity	$p = mv$
force = $\frac{\text{change in momentum}}{\text{time}}$	$F = \frac{\Delta p}{t}$
u = initial velocity v = final velocity t = time a = acceleration x = displacement	$v = u + at$ $x = \frac{u + v}{2} t$
moment = force × distance	$M = Fd$

SI multipliers

Prefix	Multiplier
m	1×10^{-3}
k	1×10^{3}
M	1×10^{6}

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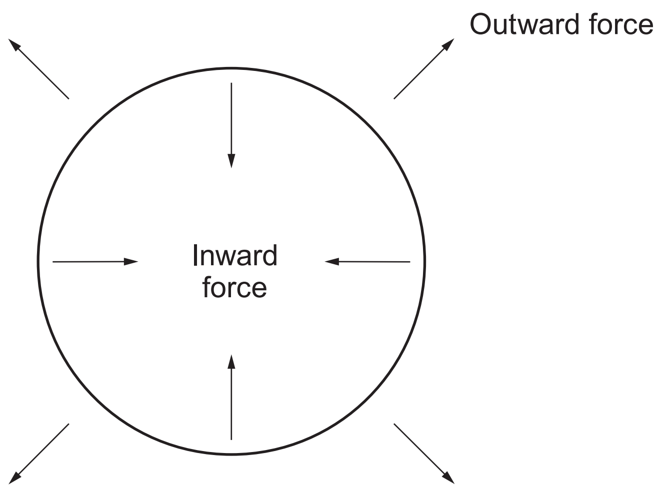
Answer **all** questions.

1. The diagram shows the forces acting on a large star.



(a) Complete the sentences by underlining the correct word in brackets.

(i) The inward force acting on the star is caused by (**gravity** / **pressure** / **helium**). [1]

(ii) The outward force acting on the star is caused by (**gravity** / **pressure** / **helium**). [1]

(b) Use only words from the box to answer the following questions.

space,	**white dwarf,**	**supergiant,**	**neutron star,**	**solar system**

(i) Complete the list of stages that this large star will go through, from the main sequence until it reaches the end of its life. [2]

main sequence star → .... → supernova → ....

(ii) Complete the following sentence using words from the box. [2]

During the supernova stage, material is returned to ....

where it creates a new ....

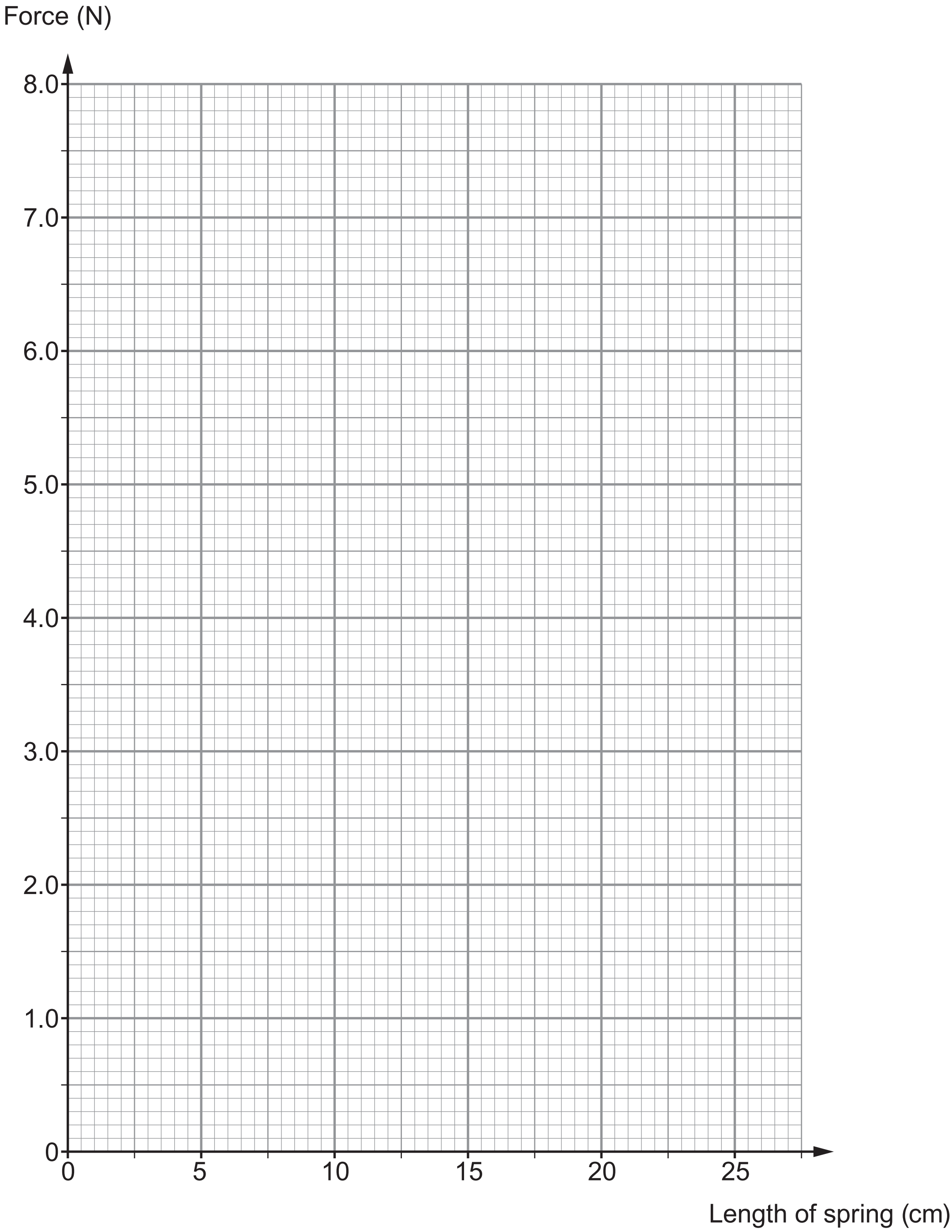
.

6

2. The following results were obtained from an investigation into stretching a spring.

Force (N)	Length of spring (cm)
1.0	12.0
2.5	15.0
4.0	18.0
5.0	20.0
6.0	22.0
7.5	25.0

(a) Plot the data on the grid below and draw a suitable straight line. [3]



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(b) Use the graph to answer the following questions.

(i) The unstretched length of the spring is its length when the force applied is zero. State the unstretched length of the spring. [1]

Unstretched length = ... cm

(ii) I. State the length of the spring produced by a force of 3.0 N. [1]

Length = ... cm

II. Use the equation:

extension = length produced by 3.0 N – unstretched length

to calculate the extension produced by a force of 3.0 N. [1]

Extension = ... cm

(iii) Use your previous answer and the equation:

$$\text{spring constant} = \frac{\text{force}}{\text{extension}}$$

to calculate the spring constant of the spring for a force of 3.0 N. [2]

Spring constant = ... N/cm

(iv) Complete the sentence by underlining the correct answer in the brackets. [1]

If the force on the spring is doubled to 6.0 N, then the spring constant (**halves** / **stays the same** / **doubles**).

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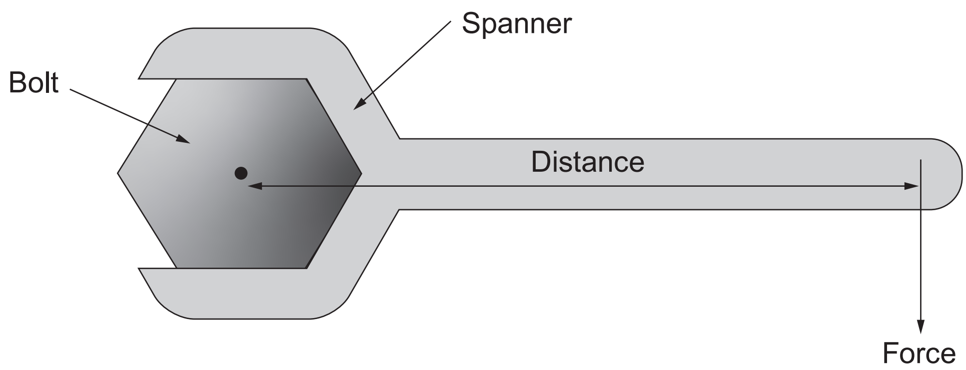
3. The diagram shows a spanner being used to tighten a bolt.



A force at the end of the spanner produces a moment about the bolt.

(a) In the diagram above, the force is 80 N and the distance is 0·25 m.

Use the equation:

$$\text{moment} = \text{force} \times \text{distance}$$

to calculate the moment of the force about the bolt. [2]

Moment = ... N m

(b) Henry says it is easier to tighten the bolt if a longer spanner is used.
Explain why Henry is correct. [2]

....

....

....

4

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4. This question is about nuclear fusion in a star.

(a) State the **two** conditions that are required for nuclear fusion to take place. [2]

1. ....

2. ....

(b) Deuterium and tritium are isotopes of hydrogen.
In one nuclear fusion reaction, deuterium and tritium undergo nuclear fusion to form helium and one neutron.

Complete the nuclear equation below for the fusion of deuterium and tritium. [3]

$$^{2}_{1}\mathrm{H} + ^{3}_{1}\mathrm{H} \longrightarrow ^{\dots\dots}_{\dots\dots}\mathrm{He} + ^{1}_{\dots\dots}\mathrm{n}$$

5

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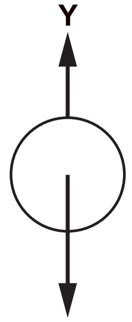
5. A ball is dropped from rest.
Two forces, X and Y, act on the ball during its fall.



(a) (i) Name the forces X and Y. [2]

X .... Y ....

(ii) State what happens to the size of the forces, if anything, as the ball speeds up. [2]

Force X ....

Force Y ....

(b) Complete the following sentences by underlining the correct phrase in the brackets. [2]

(i) The ball speeds up at the start of its fall because force Y is (**smaller than** / **equal to** / **larger than**) force X.

(ii) The ball eventually reaches a constant speed because force Y is (**smaller than** / **equal to** / **larger than**) force X.

(c) After being dropped from rest, the ball takes 2 seconds to hit the ground.

(i) Use the equation:

$$v = u + at$$

to calculate the final speed, v, with which the ball hits the ground. [2]
(Acceleration due to gravity, $g = 10\,m/s^2$)

v = ... m/s

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(ii) Use an equation from page 2 and your answer to part (c)(i) to calculate the height, x, from which the ball was dropped. [3]

x = ... m

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6. A group of students use 6-sided dice to model radioactive decay.
There are 240 dice in total.
Dice with a 6 facing upward represent decayed nuclei.

(a) (i) State what the students should do after throwing the 240 dice for the first time. [1]

....

....

(ii) Describe how they should continue the experiment. [1]

....

....

(iii) I. State the chance (probability) of any one of the dice landing with a 6 facing upward. [1]

....

II. Use your previous answer to predict how many dice will land with a 6 facing up on the first throw. [1]

....

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(b) The table shows the results.

Throw	Number of sixes	Number of dice remaining
0	0	240
1	38	202
2	30	....
3	28	144
4	22	122
5	22	100

(i) **Complete** the table. [1]

(ii) State how many dice remain after one half-life. .... [1]

(iii) Use the data in the table to estimate the half-life of the 'dice' in throws. [1]

Half-life = ... throws

7

Examiner only

7. (a) In dry conditions, a Formula One (F1) car can accelerate from 0 to 30 m/s in 1.5 seconds in a straight line.

(i) State the change in velocity. [1]

Change in velocity = ... m/s

(ii) Use the equation:

$$\text{acceleration} = \frac{\text{change in velocity}}{\text{time}}$$

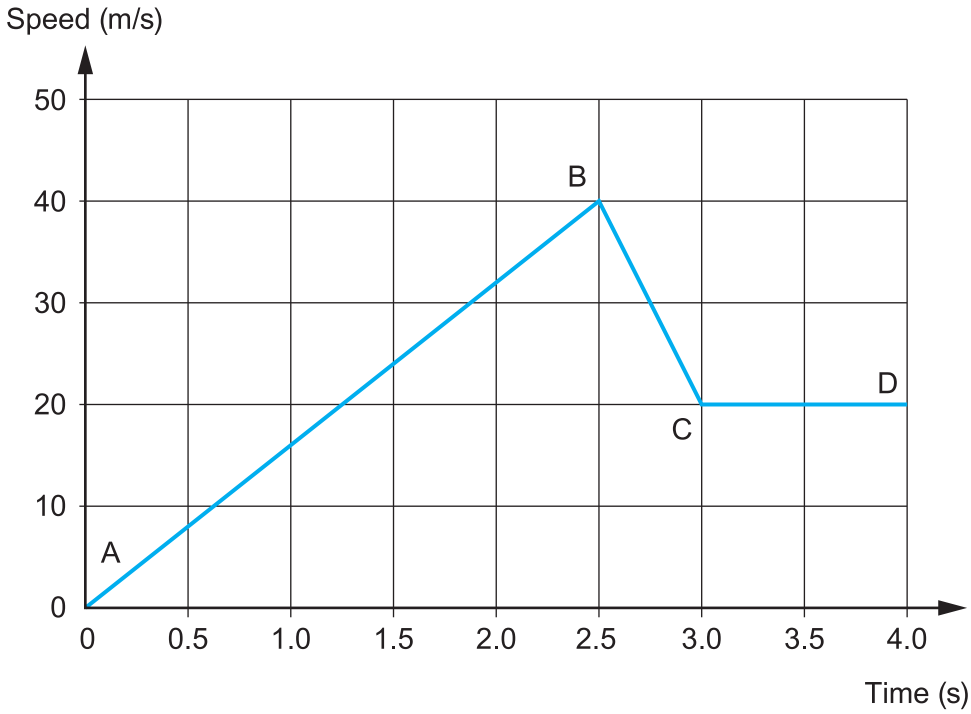
to calculate the acceleration of the racing car. [2]

Acceleration = ... m/s^2

(b) The photograph shows F1 cars lined up on the grid.

Examiner only

The graph below shows how the speed of a F1 racing car changes at the start of a race as it leaves the grid and goes around the first bend.



(i) **Use data from the graph** to describe the motion of the car during the time shown. No calculations are required. [6 QER]

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(ii) The car travels 85 m during the time shown by the graph.
Use an equation from page 2 to calculate the mean speed of the car during this time. [3]

Mean speed = ... m/s

12

Examiner only

8. Astronomers have studied light from distant galaxies.
They found that the light from these galaxies has increased in wavelength.

(a) Complete each sentence below by **drawing a line** to the correct ending. [2]

(i)	The increase in wavelength of light is due to the Universe…	contracting staying the same size expanding
(ii)	The increase in wavelength of light supports the Big Bang model which states the Universe started from…	different points a single point a supernova

(b) Look at the diagrams below.
They show absorption spectra of hydrogen in a laboratory on Earth (**Diagram A**), from a distant galaxy X (**Diagram B**) and from another distant galaxy Y (**Diagram C**).

Diagram A: Laboratory on Earth

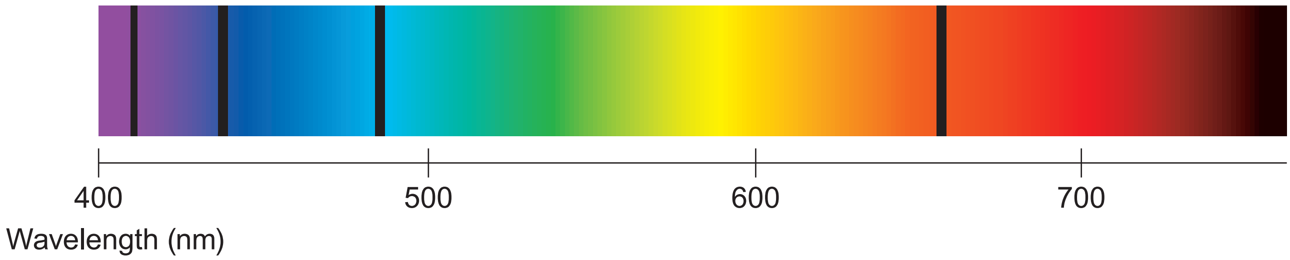


Diagram B: From galaxy X

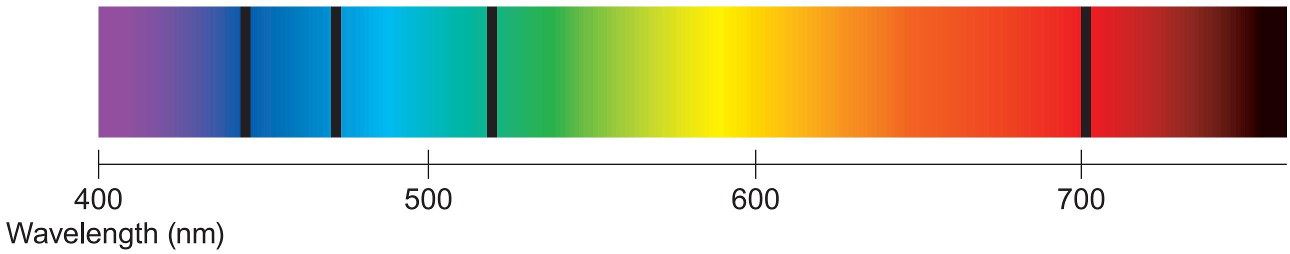
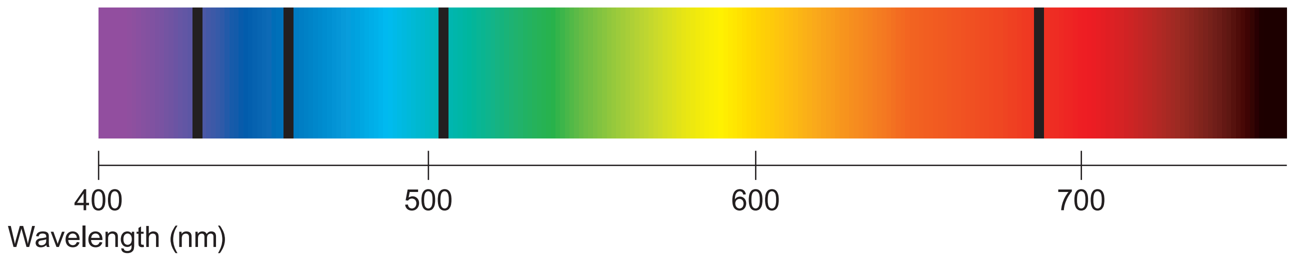


Diagram C: From galaxy Y



Examiner only

Use the spectra opposite to answer the following questions.

(i) Compare the spectral dark lines in diagrams **A** and **B**. [2]

....

....

....

(ii) State the **differences** between the speed **and** distance from Earth for galaxies **X** and **Y**. [2]

....

....

....

6

Examiner only

9. (a) Background radiation occurs due to the decay of an unstable nucleus releasing alpha, beta or gamma radiation.

Complete the table below to state what each type of radiation is. [3]

Type of radiation	Symbol	What it is
Alpha	${}^{4}_{2}\alpha$	....
Beta	${}^{0}_{-1}\beta$	....
Gamma	γ	....

(b) A teacher demonstrates how to determine the background radiation count in her laboratory.
She uses a radiation detector and measures 18 counts in 30 seconds.

(i) Determine the background radiation count in counts per second. [1]

Background radiation = ... counts per second

(ii) Suggest **two** ways that the teacher could improve her measurement of background radiation. [2]

1. ....

2. ....

(iii) Explain why the background radiation count is much higher in some parts of the country than others. [2]

....

....

....

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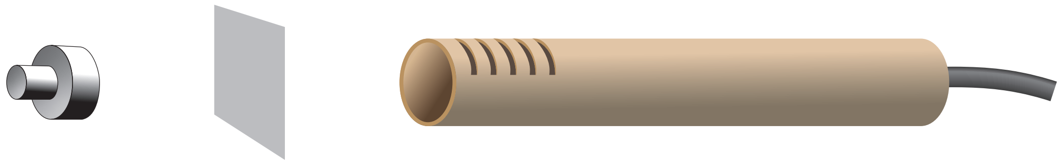
(c) The teacher now demonstrates an experiment to identify the radiation emitted by different sources.
The count rate is measured, first with no absorber present and then with paper and aluminium absorbers separately.



The results are shown in the table below. They are corrected for background radiation.

Source	Count rate (counts per second)		
	No absorber	Paper	Thin aluminium
1	312	313	312
2	389	57	0

(i) Write **yes (Y) or no (N)** in each box to show which type(s) of radiation is (are) emitted by each source. [4]

	Alpha	Beta	Gamma
Source 1			
Source 2			

(ii) Explain **one** change the teacher could make to extend the investigation to confirm to the class the conclusions made. [2]

....

....

....

14

Examiner only

10. The table below gives data about some objects in our solar system.

Object	Mass (units)	Diameter (km)	Length of day (hours)	Year length (days)	Orbital speed (km/s)	Mean temperature (°C)	Distance from Sun (units)
Mercury	0.330	4879	4222.6	88	47.4	167	0.39
Venus	4.87	12104	2802	225	35	464	0.72
Earth	5.97	12756	24	365	29.8	15	1.00
Moon	0.073	3475	708.7		1	–20	
Mars	0.642	6792	24.7	687	24.1	–65	1.52
Jupiter	1898	142984	9.9	4331	13.1	–110	5.20
Saturn	568	120536	10.7	10747	9.7	–140	9.54
Uranus	86.8	51118	17.2	30589	6.8	–195	19.18
Neptune	102	49528	16.1	59800	5.4	–200	30.06
Pluto	0.0146	2370	163.3	90560	4.7	–225	39.53

(a) (i) Use the information from the table to **tick (✓)** the **two** correct statements below. [2]

Neptune is hotter than the Moon. ☐

The mean temperature of the Moon is 5 degrees less than the Earth. ☐

A year on Earth is about 4 times longer than a year on Mercury. ☐

Mercury orbits the Sun with a speed around 10 times greater than Pluto. ☐

(ii) Pluto was once considered to be a planet but is now classed as a dwarf planet. Use the data in the table to suggest a reason for the change. [1]

....

....

Examiner only

(iii) Ceres is another dwarf planet and it is the only dwarf planet located in the asteroid belt. Estimate its distance from the Sun. [1]

Distance from the Sun = ... units

(b) Elin concludes that rocky planets with the greatest mass have the shortest day length. Explain the extent to which the data supports this conclusion. [2]

....

....

....

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END OF PAPER

Surname	Centre Number	Candidate Number
First name(s)		0

GCSE

3420U20-1

S23-3420U20-1

THURSDAY, 25 MAY 2023 – MORNING

PHYSICS – Unit 2:
Forces, Space and Radioactivity

FOUNDATION TIER

1 hour 45 minutes

For Examiner's use only		
Question	**Maximum Mark**	**Mark Awarded**
1.	**7**	
2.	**7**	
3.	**8**	
4.	**9**	
5.	**9**	
6.	**14**	
7.	**6**	
8.	**7**	
9.	**13**	
Total	**80**	

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01

ADDITIONAL MATERIALS

In addition to this paper you will require a calculator and a ruler.

INSTRUCTIONS TO CANDIDATES

Use black ink or black ball-point pen. Do not use gel pen or correction fluid.

You may use a pencil for graphs and diagrams only.

Write your name, centre number and candidate number in the spaces at the top of this page.

Answer **all** questions.

Write your answers in the spaces provided in this booklet. If you run out of space use the additional page at the back of the booklet, taking care to number the question(s) correctly.

INFORMATION FOR CANDIDATES

The number of marks is given in brackets at the end of each question or part-question.

The assessment of the quality of extended response (QER) will take place in question **7**.

Equations

speed = $\dfrac{\text{distance}}{\text{time}}$	
acceleration [or deceleration] = $\dfrac{\text{change in velocity}}{\text{time}}$	$a = \dfrac{\Delta v}{t}$
acceleration = gradient of a velocity-time graph	
resultant force = mass × acceleration	$F = ma$
weight = mass × gravitational field strength	$W = mg$
work = force × distance	$W = Fd$
force = spring constant × extension	$F = kx$
momentum = mass × velocity	$p = mv$
force = $\dfrac{\text{change in momentum}}{\text{time}}$	$F = \dfrac{\Delta p}{t}$
u = initial velocity v = final velocity t = time a = acceleration x = displacement	$v = u + at$ $x = \dfrac{u + v}{2}\,t$
moment = force × distance	$M = Fd$

SI multipliers

Prefix	Symbol	Conversion factor	Multiplier
milli	m	divide by 1000	1×10^{-3}
centi	c	divide by 100	1×10^{-2}
kilo	k	multiply by 1000	1×10^{3}
mega	M	multiply by 1 000 000	1×10^{6}

Examiner only

Answer **all** questions.

1. (a) The table below contains statements about the motion of an object and the forces acting on it.
For each statement, place **one** tick (✓) to show which law it applies to.
One row has been completed as an example. [3]

Statement	Newton's 1st Law	Newton's 2nd Law	Newton's 3rd Law
A skydiver accelerates when the forces are unbalanced.	✓		
This law can be written as $F = ma$.			
A train will not move from rest unless acted on by a resultant force.			
When it is fired, a rifle exerts a force on a bullet. The bullet exerts an equal and opposite force on the rifle.			

(b) A ball is dropped from rest (u = 0 m/s) from a window.
It takes a time, t = 1.2 s to reach the ground.

(i) Use the equation:

$$v = u + at$$

to calculate the speed, v, of the ball as it hits the ground. [2]
(Acceleration, a = 10 m/s^2)

v = ... m/s

(ii) Use your answer above for v and the equation:

$$x = \frac{u + v}{2}t$$

to calculate the distance, x, of the window above the ground. [2]

x = ... m

7

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03

Examiner only

2. Caesium has many isotopes.
One of these is caesium-137.

(a) The symbol for the isotope caesium-137 is $^{137}_{55}\text{Cs}$.

Tick (✓) the box next to the symbol of another isotope of caesium. [1]

$^{137}_{54}\text{Cs}$ ☐

$^{133}_{55}\text{Cs}$ ☐

$^{138}_{57}\text{Cs}$ ☐

(b) When caesium-137 decays, beta particles are emitted from unstable nuclei.

(i) Tick (✓) the box next to the correct statement about unstable nuclei. [1]

They have too many electrons ☐

They have too many protons ☐

The number of neutrons and protons is unbalanced ☐

(ii) Tick (✓) the box next to the correct symbol for a beta particle. [1]

$^{1}_{-1}\beta$ ☐

$^{0}_{-1}\beta$ ☐

$^{-1}_{0}\beta$ ☐

(iii) Tick (✓) the box next to the correct statement about a beta particle. [1]

It is a helium nucleus ☐

It is an electromagnetic wave ☐

It is a high energy electron ☐

Examiner only

(c) The activity of a sample of caesium-137 is 160 units.
It has a half-life of 30 years.

(i) Underline **one** number in the brackets to complete the sentence. [1]

After 30 years, the activity of the caesium-137 sample will be (**20 / 40 / 80**) units.

(ii) Jason says the activity of the sample will be zero after 60 years.
Explain whether you agree with Jason. [2]

....

....

....

7

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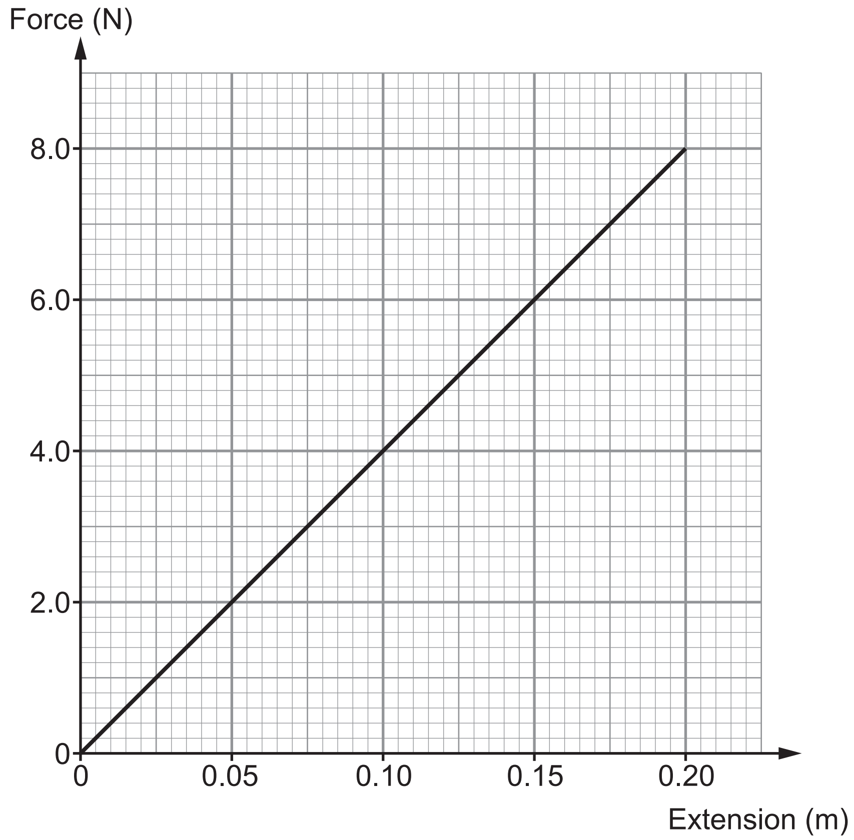
3. (a) Students carry out an experiment on a spring.
The length of the spring is measured for each force and extensions are calculated.
The force-extension graph is shown below.



Use the equation:

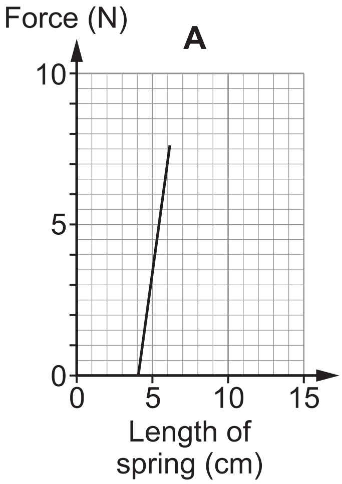
$$\text{spring constant} = \frac{\text{force}}{\text{extension}}$$

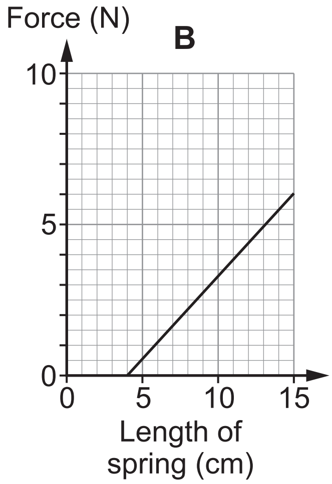
and the graph to calculate the spring constant in N/m for a force of 8.0 N. [2]

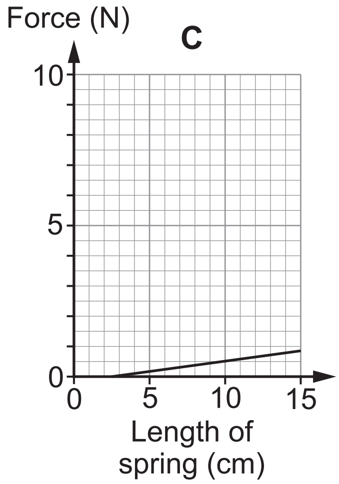
spring constant = ... N/m

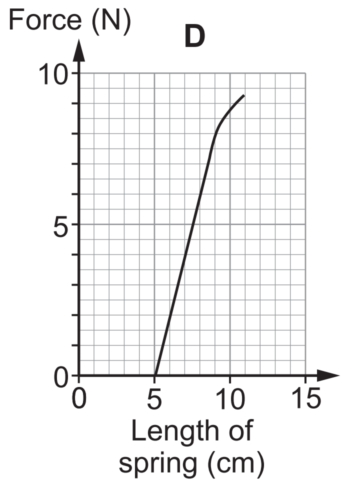
Examiner only

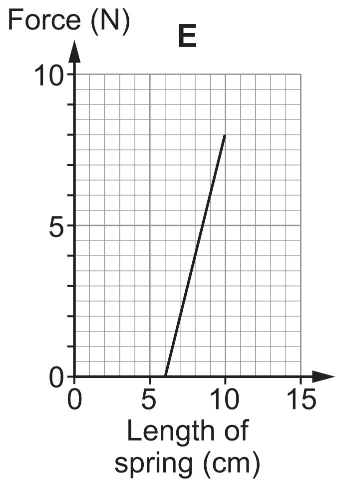
(b) The following graphs were obtained when students in a class used forces to stretch five different springs, **A**, **B**, **C**, **D** and **E**.
These students measured the lengths of each spring with different forces and did not calculate extension.
All graphs are drawn on axes with the same scales.











(i) Use the graphs above to answer the following questions about springs **A** to **E**.

I. State which spring is easiest to stretch. .... [1]

II. State which spring has the biggest unstretched length. .... [1]

III. State which spring has the same spring constant as **D**. .... [1]

IV. State which spring is stretched beyond its elastic limit. .... [1]

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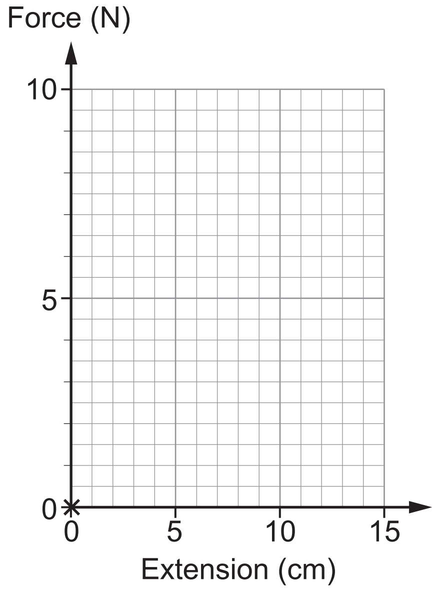
(ii) They decided to calculate extensions and draw the **force-extension** graph for Spring **E**.
Draw this graph on the axes below.
The first point has been plotted for you. [2]



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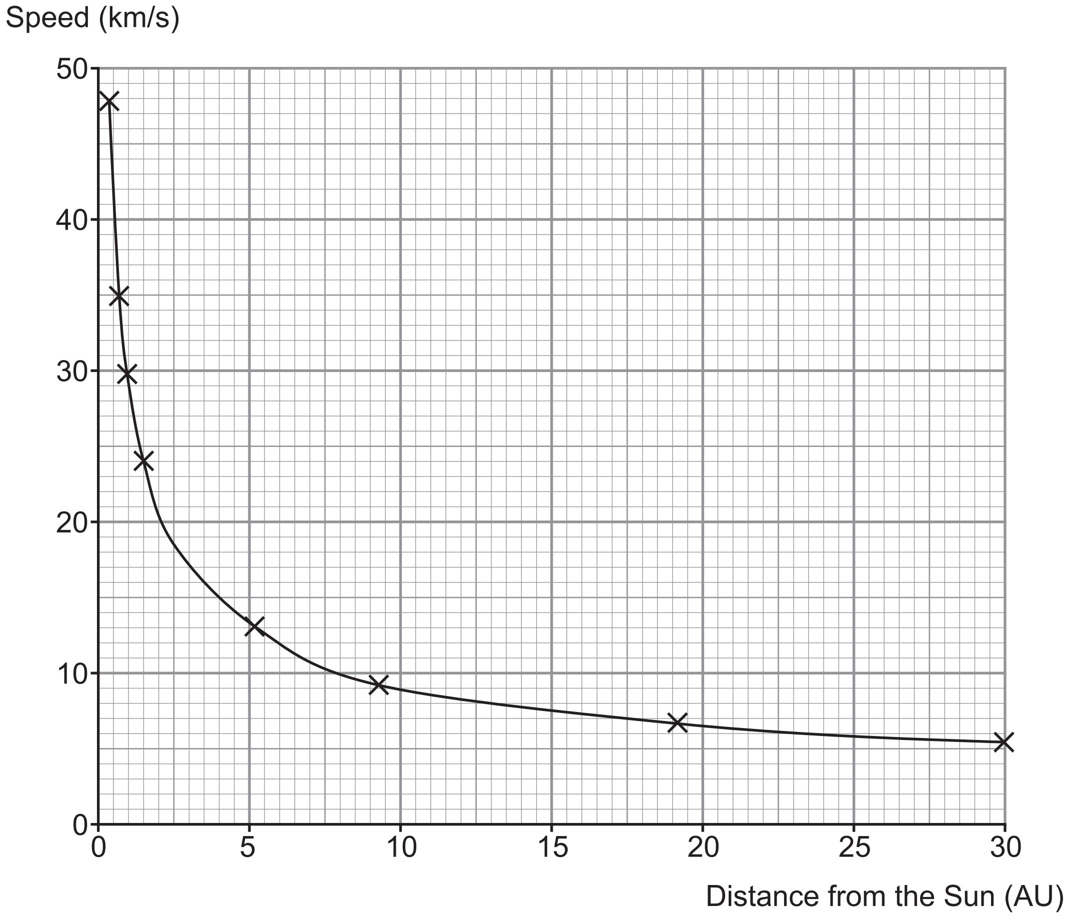
4. There are eight planets in orbit around the Sun in our solar system.
The graph shows how their orbital speeds depend on their distance from the Sun.



(a) Describe the relationship between the speed of the planets and their distance from the Sun. [2]

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Examiner only

(b) Use the information in the graph to **complete the table**. [2]

Planet	Mercury	Venus	Earth	Mars	Jupiter	Saturn	Uranus	Neptune
Speed (km/s)	47.9	....	29.8	24.1	13	9.7	6.8	5.4
Distance from the Sun (AU)	0.4	0.7	1	1.5	5.2	9.5	19.2	....

(c) Name the planet that is about twice as far as Saturn from the Sun. [1]

....

(d) The distances on the graph are given in AU (astronomical units).
Use the table to state what distance one AU represents. [1]

....

(e) The asteroid belt is between the orbits of Mars and Jupiter.
Paula states that the mean orbital speed of asteroids is 18 km/s.
Owain states that the asteroid belt is 10 AU from the Sun.
Use the data in the table above to explain whether you agree with Paula or with Owain. [3]

....

....

....

....

....

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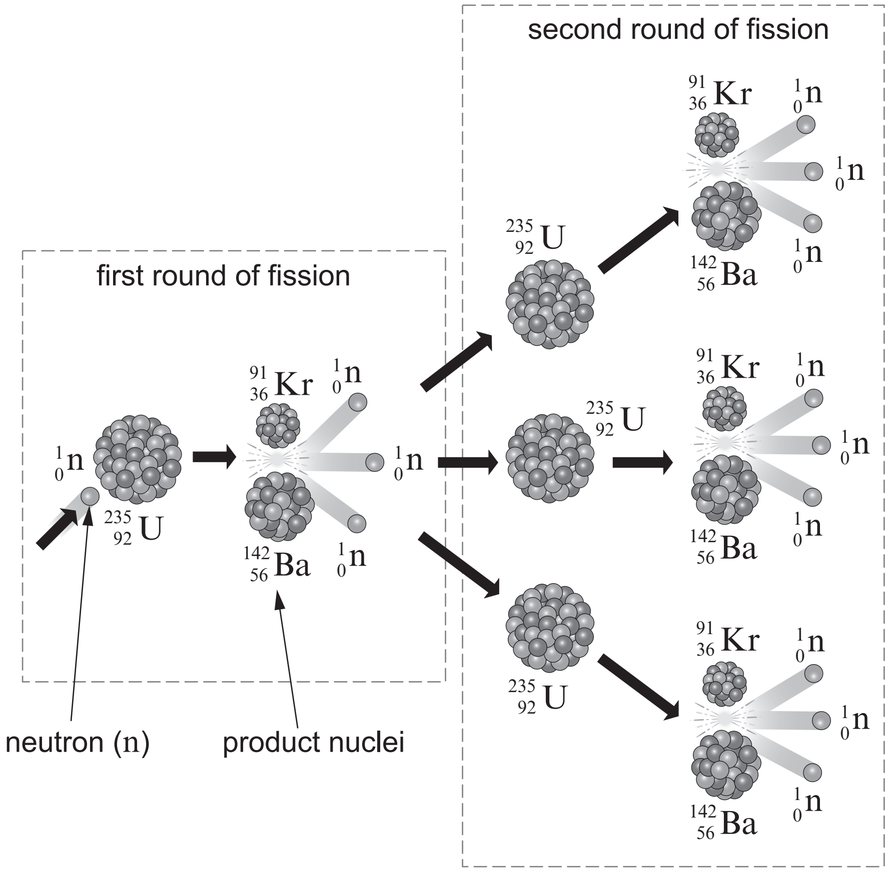
9

Examiner only

5. The diagram shows one possible chain reaction in the fission of uranium–235 ($^{235}_{92}U$) by a single neutron ($^{1}_{0}n$).
The product nuclei, krypton (Kr) and barium (Ba), are radioactive.



During the first round of fission, 2 product nuclei are created and 3 neutrons are released.

(a) Tick (✓) the **three** boxes next to the correct statements below. [3]

One neutron splits to create three other neutrons. ☐

The proton numbers of barium and krypton add to give the proton number of uranium-235. ☐

Nuclear fission takes in energy from the nucleus. ☐

During the second round of fission, another 9 neutrons are released. ☐

During the second round of fission, another 6 product nuclei are created. ☐

An atom of uranium-235 has 92 neutrons in its nucleus. ☐

Examiner only

(b) Use words from the box to complete the following sentences about a nuclear reactor. Each word may be used once, more than once or not at all.

absorb	**one**	**two**	**three**	**fuel**	**moderator**
emit	**concrete**	**reactor**	**control**		

(i) Neutrons are slowed down by the .... so uranium nuclei can .... them. [2]

(ii) .... rods absorb fission neutrons so only .... of these neutrons goes on to produce further fission. [2]

(iii) In an emergency, ... rods are dropped fully into the nuclear .... [2]

9

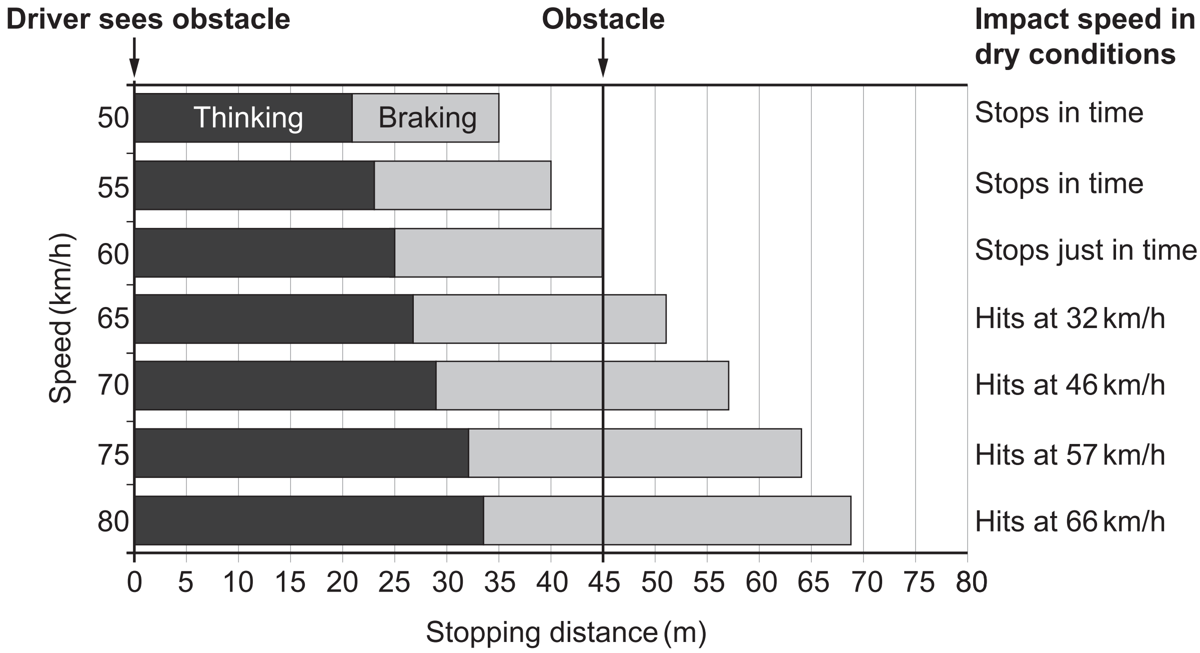
Examiner only

6. The chart below is used by traffic collision investigators. It gives the thinking, braking and stopping distances of cars driven at different speeds by an alert driver on a dry road.

Stopping distance is given by the following equation:

stopping distance = thinking distance + braking distance

An alert driver notices an obstacle 45 m away on the road ahead. The position of this obstacle is represented by the dark vertical line. If there is a collision, the chart also shows the impact speed with the obstacle.



(a) Use the information in the chart to answer the following questions.

(i) State the stopping distance for a speed of 50 km/h. .... m [1]

(ii) State the speed at which the car stops just in time. .... km/h [1]

(iii) State the speed which gives a braking distance of 35 m. .... km/h [1]

Examiner only

(iv) Tiredness doubles **thinking** distance for some drivers.
Gareth claims that, for these tired drivers travelling at 60 km/h, the stopping distance becomes 90 m.
With the aid of calculations, explain whether you agree with the claim. [3]

....

....

(v) Use the equation:

$$\text{time} = \frac{\text{distance}}{\text{speed}}$$

and information from the chart for a car travelling at 60 km/h (17 m/s), to calculate the **thinking time** of an alert driver. [3]

Thinking time = ... s

(b) A car is travelling at 70 km/h on a dry road when it starts to rain causing the road to become **wet**.
Complete the table below.
In each box, add either increases, decreases, or stays the same. [3]

Thinking distance	Braking distance	Stopping distance	Impact speed
....	....	increases	....

Examiner only

(c) Seat belts **and** crumple zones work together to keep the occupants of a car safe in the event of a head-on collision.
Complete the table by placing a tick (✓) in the column that matches with the action.
One has been done as an example. [2]

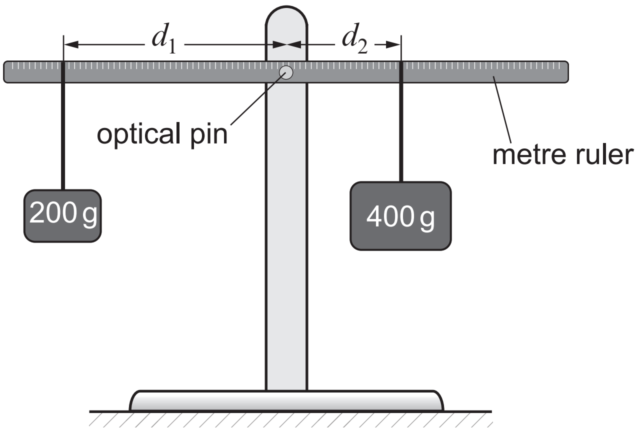
Action	Seat belt	Crumple zone
Increases the time of the collision		✓
Reduces force on the car		
Prevents driver continuing through the windscreen		

14

Examiner only

7. Describe how you would use the apparatus listed below and shown in the diagram to investigate the principle of moments. [6 QER]

List of apparatus:
- metre ruler with small hole at centre
- 2 × 100 g mass hangers
- 8 × 100 g masses
- 2 × loops of cotton
- clamp stand, boss and clamp
- optical pin and cork
- small piece of plasticine



Include in your answer:
- How you will set up the apparatus
- What measurements you will take
- How you will analyse your results to show the principle of moments.

....

....

....

....

....

....

....

....

....

....

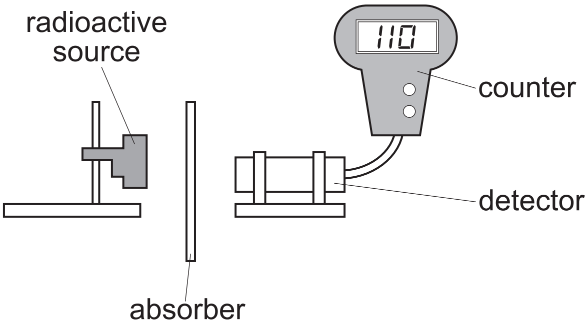
....

Examiner only

6

Examiner only

8. A teacher uses the apparatus below to demonstrate the penetrating properties of alpha, beta and gamma radiation.



(a) The teacher explains that there is a possibility of exposure to radiation from the source. Complete the risk assessment below. [2]

Hazard	Risk	Control measure
Nuclear radiation is ionising	....	....

(b) After the experiment the teacher gives the students some data about the radioactive source, cobalt-60, to analyse.
The data are given in the table below.

Absorber	Count rate (counts per second)
no absorber	256
paper	256
aluminium	110
lead	50

Use the data to answer the following questions.

(i) Explain how the data show that cobalt-60 does not emit alpha particles. [1]

....

....

Examiner only

(ii) Explain how the data show that cobalt-60 emits beta and gamma radiation. [2]

....

....

....

(iii) The teacher tells the class that counts due to background radiation are included in the results in the table.

I. State **one** cause of background radiation. [1]

....

II. State how the results in the table should be corrected for background radiation. [1]

....

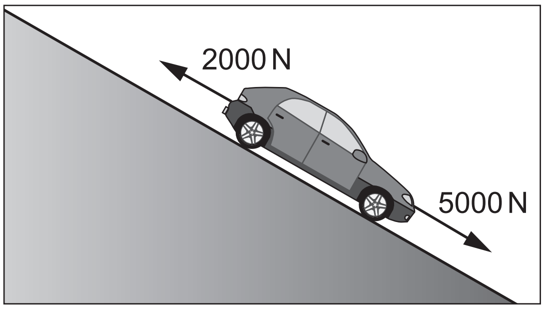
....

....

7

Examiner only

9. The diagram shows a car rolling down a slope. Two of the forces acting on the car are labelled.



(a) The car has a **weight** of 10 000 N. Use the equation:

$$\text{mass} = \frac{\text{weight}}{\text{gravitational field strength}}$$

to calculate the mass of the car. [2]
(Gravitational field strength, g, = 10 N/kg)

mass = ... kg

(b) Use the information in the diagram to answer the following questions.

(i) Calculate the resultant force acting down the slope. [2]

resultant force = ... N

(ii) Use your answers from parts (a) and (b)(i) and the equation:

$$\text{acceleration} = \frac{\text{resultant force}}{\text{mass}}$$

to calculate the acceleration of the car at the instant shown and state the unit. [3]

acceleration = ...

unit = ...

Examiner only

(iii) I. Explain how the resultant force on the car changes as it speeds up. [2]

....

....

....

II. State how this change in resultant force affects the acceleration of the car. [1]

....

(c) At the bottom of the slope the car continues horizontally at a constant speed of 12 m/s with a kinetic energy of 72 000 J.

(i) State **one** reason why the potential energy at the top of the hill must have been greater than 72 000 J. [1]

....

(ii) At the bottom of the hill a braking force is applied which stops the car over a distance of 15 m.

Use the equation:

$$\text{force} = \frac{\text{work done}}{\text{distance}}$$

to calculate the braking force. [2]

braking force = ... N

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END OF PAPER